PHYSICS GRADE 10: PRESSURE

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Physics
Strand	Strand 1.0: Mechanics and Thermal Physics
Sub-Strand	Sub-Strand 1.2: Pressure
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Atmospheric pressure and its effects – Factors affecting pressure in liquids (depth, density of liquid, gravitational field strength) – Application of the equation P = ρgh to determine pressure in liquids – Transmission of pressure in fluids (Pascal's Principle) – Applications of pressure in fluids: drinking straw, syringe, hydraulic lift, hydraulic brakes – Pressure in gases: Boyle's Law and the relationship between pressure and volume – Simple mercury and aneroid barometers – Measurement of pressure using manometers – Real-life applications of pressure: blood pressure, pneumatic systems, water supply systems
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe atmospheric pressure and explain its effects on everyday objects and phenomena, b) identify and explain the factors that affect pressure in liquids, c) apply the equation P = ρgh to solve problems involving pressure in liquids, d) explain the transmission of pressure in fluids using Pascal's Principle, e) describe applications of pressure in fluids including the hydraulic lift, hydraulic brakes, syringe, and drinking straw, f) investigate the relationship between pressure and volume of a gas (Boyle's Law), g) describe how barometers and manometers are used to measure atmospheric and gas pressure, h) appreciate the importance of understanding pressure in solving real-life problems in Kenyan contexts such as water supply, medicine, and engineering.
Core Competencies	– Communication and Collaboration: Learners work in groups to design and conduct experiments on liquid pressure and gas pressure, share findings with the class, and collaboratively build pressure models — developing teamwork and clear scientific communication. – Critical Thinking and Problem Solving: Learners apply P = ρgh and Boyle's Law to solve structured problems set in Kenyan contexts (e.g., calculating water pressure at the base of a Nairobi water tower), and evaluate the design of hydraulic systems. – Learning to Learn: Learners take ownership of their understanding by updating their Driving Question Board across lessons, reflecting on how each new concept revises their earlier predictions about the anchoring phenomenon. – Digital Literacy: Learners use digital simulations or videos (where available) to visualise gas particle behaviour under changing

	pressure and to explore virtual hydraulic systems.
Core Values	– Responsibility: Learners take responsibility for safe and accurate conduct of pressure experiments, handling mercury (if used) or syringes carefully, and recording data honestly. – Respect: Learners demonstrate respect for peers' ideas and predictions during group discussions and DQB sessions, appreciating diverse approaches to explaining pressure phenomena. – Unity: Learners collaborate across gender and ability groups when conducting experiments, reinforcing that scientific investigation is a shared community endeavour relevant to all Kenyans.
Science & Engineering Practices	– SEP 1 – Asking Questions and Defining Problems: Learners generate questions for the Driving Question Board after observing the anchoring phenomenon of a crushed plastic bottle, identifying what they need to understand about atmospheric pressure. – SEP 2 – Developing and Using Models: Learners build and revise particle-level and systems-level models of pressure in solids, liquids, and gases across the unit's Model Building Phase in each lesson. – SEP 3 – Planning and Carrying Out Investigations: Learners design controlled experiments to investigate how depth and liquid density affect pressure, and to verify Boyle's Law using a sealed syringe and weights. – SEP 4 – Analysing and Interpreting Data: Learners plot P vs. 1/V graphs from Boyle's Law data and P vs. depth graphs from liquid pressure investigations, interpreting gradients in terms of ρg. – SEP 6 – Constructing Explanations: Learners construct written and verbal explanations for why a straw works, why a hydraulic jack can lift a matatu, and why deep-sea fish die when brought to the surface near Lake Victoria. – SEP 8 – Obtaining, Evaluating, and Communicating Information: Learners research and present on one real-world Kenyan application of fluid pressure (e.g., Nairobi City Water and Sewerage Company water towers, hospital drip systems, sugarcane press hydraulics in western Kenya).
Pertinent & Contemporary Issues (PCIs)	– Environmental Education: Learners explore how changes in atmospheric pressure relate to weather patterns relevant to Kenyan farmers in the Rift Valley and coastal Mombasa, linking pressure science to climate awareness. – Health Education: Learners examine how blood pressure monitoring (a direct application of fluid pressure) is used in Kenyan hospitals and community health clinics, connecting physics to personal and public health. – Gender Disparity: Learners critically examine and challenge stereotypes that portray Physics — and engineering careers built on pressure principles — as male-dominated fields, by highlighting contributions of female Kenyan engineers and scientists.
Career Connections	– Civil and Hydraulic Engineering: Understanding $P = \rho gh$ and Pascal's Principle is foundational for designing water supply systems, dams (e.g., Masinga Dam), and irrigation networks across Kenya. – Medicine and Nursing: Blood pressure measurement, intravenous drip systems, and syringe design are all direct applications of fluid pressure studied in this sub-strand. – Aviation and Meteorology: Atmospheric pressure and barometry underpin weather forecasting at Kenya Meteorological Department stations and flight instrument calibration at JKIA. – Automotive and Industrial Technology: Hydraulic brakes and lifts studied in this unit are core technologies used in Kenyan garage mechanics, the Mombasa port, and manufacturing plants in Athi River's Export Processing Zone. – Agriculture and Water Resources: Farmers and water engineers in Kericho and the Rift Valley use pressure principles to design drip irrigation and gravity-fed water systems.
Focus for Lessons	This unit anchors learning in the puzzling everyday observation that the atmosphere — though invisible — exerts enormous force on all objects, and that fluids transmit pressure in ways that enable both life (blood circulation, drinking through a straw) and powerful machines (hydraulic jacks used in Kenyan garages and sugarcane presses). Students move from qualitative observation of atmospheric and liquid pressure effects, through quantitative investigation using $P = \rho gh$ and Boyle's Law, to systems-level understanding of how Pascal's Principle enables hydraulic technology. A strong Kenya context is maintained throughout, with

	problems, demonstrations, and career connections drawn from Nairobi water towers, Lake Victoria fishing, Rift Valley irrigation, and hospital settings.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: "Why did the jerrycan crush itself — and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?" KICD KEY INQUIRY QUESTIONS: 1. How is pressure applied in day-to-day life? 2. How does pressure in fluids affect the design of machines and structures?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Crushed Jerrycan: A teacher brings to class a sealed, completely full 20-litre plastic jerrycan (the type ubiquitously used across Kenya to carry water). The jerrycan is then opened, the water is poured out, the empty jerrycan is resealed tightly, and students watch it do nothing unusual. The teacher then carefully removes a small amount of air from the jerrycan using a hand pump (or by briefly heating it with hot water and then sealing it), and — slowly, before the students' eyes — the jerrycan begins to crumple, crackle, and collapse inward on all sides without anyone touching it. No one is pushing it. Nothing visible is pressing on it. Yet it crushes itself. WHY IT IS PUZZLING: Students know that plastic jerrycans are strong — they carry 20 kg of water daily without deforming. So why does removing a tiny amount of air from inside cause the rigid container to collapse? What invisible force is acting? Where does it come from? Is the air outside stronger than the plastic? Why doesn't the same thing happen to our bodies? This contradiction between the jerrycan's obvious physical strength and its sudden, dramatic collapse from an invisible force creates immediate cognitive dissonance and drives genuine curiosity about the nature of atmospheric pressure, fluid pressure, and pressure transmission — the core concepts of the entire unit.
Storyline Thread	Lesson 1 — THE MYSTERY BEGINS: Anchoring Phenomenon Launch Students observe the crushing jerrycan demonstration. They record predictions, sketch force diagrams, and post their first "What do we think?" and "What are we wondering?" cards on the Driving Question Board. Introduce the concept of pressure as force per unit area; students measure pressure exerted by their own shoes on the floor (F/A) using bathroom scales and graph paper tracings. Lesson 2 — THE WEIGHT OF NOTHING: Atmospheric Pressure Students investigate evidence that the atmosphere has weight and exerts pressure in all directions — collapsing can experiment (boiling water in a tin, sealing, cooling), Magdeburg hemisphere video, and the mercury barometer diagram. Connect directly back to the jerrycan: the outside atmosphere was always pushing; removing inside air removed the balance. Students revise their DQB cards. Lesson 3 — GOING DEEPER: Factors Affecting Pressure in Liquids Students design and conduct an experiment using a tall transparent tube (or modified plastic bottles at different heights filled with water from Lake Naivasha) to show that liquid pressure increases with depth and with liquid density. Students record data in tables and identify the three factors: depth (h), density (ρ), gravitational field strength (g). Lesson 4 — THE EQUATION UNLOCKED: $P = \rho gh$ Students derive the equation $P = \rho gh$ from first principles (weight of a liquid column). They practise applying the formula to Kenyan contexts: calculating pressure at the base of the Sasumua Dam reservoir, the pressure on a diver at 10 m depth in the Indian

	Ocean off Mombasa, and the pressure at the bottom of a Nairobi water tower. Students update their models to show quantitative pressure values at different depths. Lesson 5 — PASSING THE PUSH: Pascal's Principle and Transmission of Pressure in Fluids Students use syringes connected by tubing (filled with water) to discover that pressure applied at one point is transmitted undiminished throughout the fluid. Formalise Pascal's Principle. Introduce the hydraulic press/lift concept with the $F_1/A_1 = F_2/A_2$ relationship. Students solve the garage-jack problem from the supporting phenomenon: how much input force is needed to lift a 3,000 kg matatu? Lesson 6 — MACHINES THAT USE FLUID POWER: Applications of Pressure in Fluids Students investigate and explain the working of: the drinking straw (returning to supporting phenomenon 1), the syringe and medical drip, the hydraulic brake system (demonstrated with a bicycle brake), and the hydraulic lift. Groups are assigned one application each, research its pressure principle, and present a 3-minute explanation to the class connecting it back to the jerrycan phenomenon and Pascal's Principle. Lesson 7 — SQUEEZING AIR: Boyle's Law and Pressure in Gases Students use a sealed syringe (or bicycle pump with a pressure gauge) to investigate the relationship between the volume and pressure of a fixed mass of gas at constant temperature. They record P and V data, plot P vs. V and P vs. 1/V graphs, and derive Boyle's Law (PV = constant). Connect to the collapsing mountain bottle (supporting phenomenon 4) and to why the jerrycan collapsed when air was removed. Students update their cumulative models. Lesson 8 — MEASURING PRESSURE: Barometers and Manometers Students examine the design and working of the simple mercury barometer and the aneroid barometer (used at Kenya Meteorological Department). They construct a simple water manometer from tubing to measure the pressure of a gas supply (or a lung exhalation). Students solve problems involving manometer readings and barometric height. Connect atmospheric pressure measurement to weather forecasting for Kenyan farmers. Lesson 9 — PUTTING IT ALL TOGETHER: Unit Synthesis, Model Finalisation, and DQB Resolution Students revisit the original crushed jerrycan and use the full vocabulary and equations of the unit to construct a complete, quantitative explanation of what happened and why. Each group presents their final cumulative model (particle-level + equation-level + application-level). The class resolves every question on the Driving Question Board, identifying which questions were answered, which led to new questions, and which connect to future sub-strands. Students complete the exit ticket: "What's one thing you can now explain about pressure that you couldn't before?"
Supporting Phenomena	– SUPPORTING PHENOMENON 1 — The Straw Paradox: A student sips Delmonte juice through a straw at lunch. Ask: "Are you pulling the juice up, or is something pushing it?" Nothing mechanical is pushing the juice upward — yet it rises against gravity every time. This supports inquiry into atmospheric pressure doing work on liquid surfaces. – SUPPORTING PHENOMENON 2 — The Deep-Sea Tilapia: Fishermen at Lake Victoria report that when certain deep-water fish are pulled rapidly to the surface in their nets, the fish arrive with swollen, bulging eyes or even ruptured organs. Ask: "What does the water depth have to do with what happens to a fish's body?" This supports inquiry into pressure increasing with depth ($P = \rho gh$) and the biological consequences of pressure change. – SUPPORTING PHENOMENON 3 — The Garage Hydraulic Jack: At a local Nairobi garage, a mechanic effortlessly lifts a fully loaded matatu (3,000 kg) off the ground by pressing down on a small foot pedal with one leg. Ask: "How can one person's foot-force lift a vehicle that weighs as much as 30 people?" This supports inquiry into Pascal's Principle and the mechanical

	advantage of hydraulic systems. – SUPPORTING PHENOMENON 4 — The Collapsing Mountain Bottle: A student returns from Nairobi (1,795 m elevation) to their home town at lower altitude and notices their sealed water bottle, which was flat when they bought it in Nairobi, is now tightly inflated and hard. Ask: "Why did the bottle change shape just by travelling downhill?" This supports inquiry into atmospheric pressure variation with altitude and Boyle's Law.

LESSON 1 (80 minutes): The Mystery Begins: Launching the Crushed Jerrycan Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon of the self-crushing jerrycan, elicit students' prior ideas about pressure and forces, and establish the Driving Question Board that will guide the entire unit.
Knowledge	Students will know that atmospheric pressure is a real, measurable force exerted by the air around us in all directions, and that the jerrycan collapses because the pressure outside exceeds the reduced pressure inside.
Skills	Students will practise making and recording evidence-based predictions, observing a phenomenon carefully, generating testable questions, and collaboratively organising their questions and initial ideas on a Driving Question Board.
Attitudes	Students will appreciate that invisible forces in nature are worthy of scientific investigation, and will demonstrate curiosity and open-mindedness when confronted with surprising observations.
Key Inquiry Question	How is pressure applied in day-to-day life?
Purpose in Storyline	This is the opening lesson of the entire Pressure unit. It exists to create genuine cognitive dissonance — the strong plastic jerrycan that Kenyan families trust to carry 20 kg of water every day collapses by itself when a tiny amount of air is removed. This contradiction hooks students emotionally and intellectually, surfaces their prior conceptions about force, pressure, and air, and opens the Driving Question Board that every subsequent lesson will build upon.
Safety Notes	Ensure the jerrycan used is a standard food-grade 20-litre plastic jerrycan with no cracks. If hot water is used to reduce internal air pressure, use water below 80 °C and handle with thick gloves. Do not use glass containers or metal tins for the main demonstration. Keep the hot-water basin on a stable surface away from student benches. Supervise any student handling of the hand pump.

B. LESSON OVERVIEW

Students encounter the anchoring phenomenon for the first time: a sealed, empty 20-litre plastic jerrycan slowly and dramatically crumples inward without anyone touching it. Through a structured Predict–Observe sequence, students surface their prior ideas, confront the puzzling evidence, and collectively generate the questions that will drive the next 19 lessons. The lesson closes with the creation of the class Driving Question Board (DQB). No formal definitions are introduced; the goal is productive confusion, rich questioning, and deep motivation to find out why.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Corporations and limited liability Source: Khan Academy (English	Students are shown a sealed, completely full 20-litre blue plastic jerrycan — the type used daily in homes across Nairobi, Kisumu, and Mombasa to carry water from boreholes and water kiosks. The	1. Place the full jerrycan on the bench. Say: 'Look carefully at this jerrycan. You have seen this your whole life. Your family uses it. It is strong. It carries 20 kg of water.' Pause 5 seconds. 'I am going to	Think-Pair-Share with public prediction recording. This strategy makes prior conceptions visible to the teacher and to peers. Recording predictions verbatim — without evaluation — signals that	Observable indicator: every student has written at least one sentence prediction in their book before sharing. Teacher circulates during the 3-minute silent write and notes (a) students who write

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	teacher places it on the demonstration bench and asks: 'This jerrycan carries 20 kg of water every single day without deforming. I am going to do something to it. Before I do, write down in your exercise books: What do you think will happen, and WHY?' Students write individual predictions (3 minutes, silent). They then share with a partner (Think-Pair-Share, 2 minutes). Selected pairs share with the class. The teacher records all predictions verbatim on the chalkboard under the heading 'OUR PREDICTIONS' — without correcting any of them.	do something to it. In your books, write your prediction: what will happen, and why? You have 3 minutes. Silence please.' Set a visible timer. 2. After 3 minutes, say: 'Turn to your partner. Share your prediction. You have 2 minutes.' 3. Cold-call EXACTLY 5 pairs using a random name stick. Ask each: 'What did your pair predict, and what is your reasoning?' 4. Record every prediction word-for-word on the board — do NOT say 'correct' or 'wrong'. Say only: 'Thank you — let us keep that idea.' 5. Allow 10 seconds of WAIT TIME after each student responds before calling the next. 6. Highlight any prediction that mentions 'air', 'pressure', or 'push' with a star — but still say nothing evaluative.	all ideas are worth investigating and creates a public record that students will revisit and revise throughout the unit, building metacognitive awareness of how their thinking changes.	nothing (prompt: 'Just write what you think — there is no wrong answer right now'), (b) students who use words like 'air', 'push', 'pressure', or 'force' — these students may be cold-called strategically in Phase 3 to bridge from intuition to explanation. Count: at least 5 different predictions are visible on the chalkboard before the demonstration begins.
Observe Phase 📹 **VIDEO:** Corporations and limited liability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher now runs the demonstration in three carefully staged steps, giving students time to observe and record at each step. STEP 1: The full jerrycan is opened, the water is fully poured out into a basin, and the empty jerrycan is resealed tightly. Teacher asks: 'What do you observe? Record it.' The jerrycan looks and feels normal — strong, rigid. STEP 2: The teacher slowly removes air from inside the jerrycan using a bicycle hand pump fitted with a one-way valve through the jerrycan cap (or, alternatively, pours a small amount of very hot water in, swirls it, then quickly seals the cap tightly so that steam condenses and reduces internal air). Students watch in complete silence. Over 30–90 seconds, the sides of the jerrycan begin to bow inward, creaking and crackling, until the jerrycan is	1. Say: 'I am going to do this in steps. At each step, you will record exactly what you observe — not what you think, just what you see, hear, or notice. Use the two-column table I am drawing on the board.' Draw: | WHAT I SAW / HEARD | WHAT I THINK WAS HAPPENING | 2. STEP 1 — Pour out water, reseal. Say: 'Observe. Record step 1.' Wait 30 seconds. Tap the jerrycan so students hear the hollow sound. 3. STEP 2 — Begin removing air. Say absolutely nothing. Let the collapse happen in silence. If using hot water method, say: 'Watch carefully. I am sealing it now.' Then place it on the bench. Maintain silence for the full collapse — do not narrate. Allow the sounds (crackle, creak, crunch) to do the work. 4. After the collapse is complete, say quietly: 'Record what you just observed.' Allow 60 seconds of silent writing.	Structured Observation with a Two-Column Evidence Table. Separating 'observation' from 'interpretation' is a core NGSS Science and Engineering Practice (Analysing and Interpreting Data; Planning and Carrying Out Investigations). This discipline trains students to distinguish empirical evidence from inference — a foundational scientific habit. The silence during the collapse maximises the emotional and cognitive impact of the phenomenon.	Observable indicator: every student's two-column table has at least two entries in the 'What I Saw / Heard' column that are purely observational (e.g., 'the sides bent inward', 'I heard a cracking sound', 'air rushed in when cap opened') rather than explanatory. Teacher scans books during the 60-second silent write. Red flag: a student who writes only 'air pressure crushed it' in the observation column has skipped to explanation — prompt them: 'What exactly did you see happen to the plastic? Describe the shape.'

	visibly crushed on multiple sides. STEP 3: The teacher unscrews the cap. Students hear and observe the rush of air back in, and watch the jerrycan partially recover. Students record all three observations in a two-column table: 'What I Saw / Heard / Felt' vs 'What I Think Was Happening'.	5. STEP 3 — Open the cap. Say: 'Listen.' Let students hear the air rush in. Say: 'Record what you observed.' 6. Cold-call 3 students to read ONLY their 'What I Saw / Heard' column — not their explanations yet. Say: 'Just the observations — what your senses told you. No explanations yet.' 7. WAIT TIME: 10–15 seconds after each student reads before acknowledging and moving to the next.		
Explain Phase 📹 **VIDEO:** [Corporations and limited liability] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Atmospheric Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students attempt to explain the observation using only their current knowledge — no new content is delivered by the teacher yet. In groups of four, students discuss: 'What could cause the jerrycan to crush inward when nothing visible is touching it?' Each group produces a written 'best explanation' (4–5 sentences) and a quick sketch showing any forces they think were acting on the jerrycan. Groups share their explanations. The teacher facilitates a class consensus discussion: 'Where do most groups agree? Where do we disagree?' Key ideas are recorded. The teacher then introduces just enough vocabulary to name what students have described — 'atmospheric pressure', 'pressure difference', 'force per unit area' — but does NOT yet deliver a lecture. Instead, the teacher uses a brief bridging analogy: 'Think about standing in a matatu that is very crowded. The people push on you from all sides equally — you don't fall. What would happen if all the people on your left side suddenly disappeared?' Students connect this to the jerrycan.	1. Say: 'In your groups of four, you have 6 minutes. Your job: write your group's best explanation for what crushed the jerrycan. Include a sketch with arrows showing any forces you think were acting. Go.' Set a visible timer. 2. Circulate. Listen for groups using the words 'air', 'push', 'pressure', 'outside', 'inside', 'equal'. Do NOT correct wrong ideas — say: 'Interesting — why do you think that?' 3. After 6 minutes, cold-call 4 groups (use random sticks). Each group has 60 seconds to share their explanation and sketch. 4. After all 4 groups share, ask the whole class: 'Show hands — which groups said something about air? About pushing? About inside vs outside?' Tally hands publicly. 5. Say: 'Here is a question — not an answer. Think about a crowded matatu. Everyone pushes on you equally from all sides. What would happen if suddenly all the passengers on your LEFT side disappeared — but the passengers on your right were still there?' WAIT 12 seconds. Cold-call 2 students. 6. Say: 'That is the key idea. Scientists have a name for this invisible push. It is called PRESSURE. And the air around you right now is pressing	Group Explanation with Sketch Annotation (NGSS Practice: Constructing Explanations; Developing and Using Models). Requiring a sketch with force arrows compels students to spatialise their thinking — a powerful sensemaking move that reveals whether students conceive of pressure as directional or omnidirectional. The matatu analogy uses a deeply familiar Kenyan context to build an intuitive bridge to pressure difference without replacing the cognitive work of sense-making.	Observable indicator: each group's sketch includes at least one arrow pointing inward on the jerrycan (indicating external force/pressure). Teacher notes groups whose arrows point only downward (gravity misconception) or outward — these groups need targeted questioning in the next lesson. Verbal indicator: when the matatu analogy is posed, at least 60% of students raise hands indicating they connected the analogy to the unequal pressure scenario. If fewer than 40% connect, the teacher re-poses: 'What changed inside the jerrycan when we removed air? What did NOT change outside?'

		on you — on this jerrycan — from every direction. Today we are not going to explain everything. Today our job is to ask better questions.' 7. Write on the board: 'Pressure = Force per unit area.' Say: 'We will build on this. But first — what questions does this raise for you?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Corporations and limited liability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student is given two sticky notes (or small paper squares if sticky notes are unavailable). On the FIRST sticky note, they write one question the crushed jerrycan has raised for them — something they genuinely want to know. On the SECOND sticky note, they write one connection — something in real life in Kenya that this phenomenon might be related to (e.g., 'Is this why my ears pop when I go up the Rift Valley escarpment?' or 'Is this related to how hospitals use syringes?'). Students post both notes on the class Driving Question Board — a large sheet of manila paper on the wall divided into columns: 'Questions About Air/Atmosphere', 'Questions About Liquids', 'Questions About Machines/Applications', and 'Other'. The teacher and two student volunteers cluster similar questions together. The class reads the DQB aloud together. The teacher points to the Driving Question at the top of the board and reads it aloud: 'Why did the jerrycan crush itself — and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?' The teacher says: 'Every lesson in this unit, we will return to this board. We will answer some questions. We will add new ones. By the end, you will have built the	1. Distribute two sticky notes or paper squares per student. Say: 'On your FIRST note — one question this demonstration raised for you. A question you genuinely want answered. On your SECOND note — one real-life connection in Kenya. Something you have seen, used, or experienced that might be related to what just happened. You have 3 minutes. Silence.' 2. Play quiet background music if available. Circulate and read over shoulders — note any exceptionally rich questions to highlight. 3. After 3 minutes, direct students to post their notes on the DQB in the correct column. Give them 2 minutes to post and read others' notes. 4. With two student volunteers, take 2 minutes to cluster similar questions (do this visibly, narrating: 'This question and this one seem to both be asking about air — let us put them together.'). 5. Read 6–8 of the most diverse or generative questions aloud, slowly. After each, say: 'Good — we will come back to that.' 6. Point to the Driving Question at the top of the board. Read it aloud slowly. Ask: 'Does this big question cover what you are curious about?' Cold-call 2 students for a yes/no with one-sentence reasoning. 7. Say: 'This board belongs to you. Every lesson, we add to it. Every lesson, we answer part of it. Let us begin.'	Driving Question Board Construction (a core NGSS Storyline tool for tracking growing understanding across a unit). The DQB externalises student thinking, creates collective intellectual ownership of the inquiry, and provides the teacher with a real-time map of misconceptions, gaps, and rich ideas to leverage. Sorting questions into thematic columns previews the structure of the unit without delivering content prematurely.	Observable indicator: every student has posted at least one question note and one connection note on the DQB. Teacher photographs the DQB immediately after the lesson for reference in future lessons. Quality check: scan the 'Questions About Air/Atmosphere' column — if fewer than 5 questions mention air, atmosphere, or pressure explicitly, the phenomenon has not yet generated sufficient cognitive dissonance and the teacher should revisit the demonstration briefly at the start of Lesson 2. A rich DQB will include questions such as: 'Why doesn't my body crush?', 'How deep can you dive before water crushes you?', 'How do hydraulic jacks at garages in Gikomba work?', 'Does altitude on Mt. Kenya change pressure?'

	answer together.'			
Model Building Phase 📹 **VIDEO:** Corporations and limited liability Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each student individually draws their 'Day 1 Model' — a sketch of the jerrycan before and after the air was removed, with annotated arrows showing their current best understanding of the forces at play. This is explicitly framed as a 'first draft' — not a test. Students label their model with: (a) what they think is pushing on the jerrycan, (b) from which directions, and (c) what changed when air was removed. Models are NOT collected — students keep them in their exercise books and will physically revise them with a different-colour pen after each subsequent lesson. The teacher closes by saying: 'Your model right now is your starting point. By the end of this unit, your model will look very different — and you will be able to explain not just this jerrycan, but syringes, hydraulic jacks, dam walls, and weather systems. You are scientists. The investigation starts now.'	1. Say: 'Open to a fresh page. At the top write: MY PRESSURE MODEL — LESSON 1. Draw the jerrycan before we removed air — sketch it. Now draw it after we removed air — sketch that too. On each sketch, draw arrows to show any forces you think are acting on the jerrycan. Label what each arrow represents.' 2. Allow 5 minutes of silent individual drawing. Circulate and observe — do NOT correct models. If a student is stuck, say: 'Just draw what you think. There is no wrong model today.' 3. After 5 minutes, ask 2 volunteers to hold up their books and briefly describe their model (30 seconds each). Say: 'Thank you. Notice how different our models are — that is exactly where we are supposed to be on Day 1.' 4. Say: 'Every lesson we will add to this model in a new colour. By Lesson 19, your model will be complete.' 5. Close the lesson by reading the Driving Question one final time from the DQB. Say: 'Sleep on this tonight. Ask yourself: why did the jerrycan crush itself? Tomorrow we dig deeper.' 6. Exit task: students write one sentence at the bottom of their model page: 'The one thing I am most curious about after today is ___________.'	Initial Model Construction (NGSS Practice: Developing and Using Models). Requiring students to build a model on Day 1 — before instruction — surfaces their initial mental models, creates a personal artefact they are invested in, and establishes the unit-long model revision habit. Using different ink colours per lesson makes the evolution of understanding visually explicit and builds metacognitive reflection on how learning changes thinking.	Observable indicator: every student's Day 1 model contains at least two labelled arrows (even if directionally incorrect). The exit sentence ('The one thing I am most curious about…') gives the teacher qualitative data on student engagement and curiosity. Teacher reads exit sentences before Lesson 2 and selects 3–4 to read aloud anonymously at the start of Lesson 2 to bridge continuity. Red flag: a student whose model shows no arrows at all — this student may need a one-on-one prompt at the start of Lesson 2: 'Remember how the sides bent inward? What do you think was pushing them in?'

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal: (1) Did the jerrycan collapse convincingly and dramatically enough to generate genuine surprise — or did students seem unimpressed? If the effect was weak, consider using a larger container or a more pronounced air-removal method for a repeat in Lesson 2. (2) How many distinct conceptual ideas appeared in the group explanations? Were students already using the word 'pressure', or were they reasoning purely in terms of 'pushing' and 'air'? This tells you how much prior knowledge the class brings and how quickly you can move toward formalisation in Lesson 3. (3) Look at the DQB: are students' questions mostly about the phenomenon itself, or have they already connected outward to applications (syringes, hydraulic jacks, diving, altitude)? A class that is already connecting outward is ready for a faster-paced unit. A class whose questions are mostly 'why did the jerrycan crush?' needs more phenomenon-anchoring time in Lessons 2–3. (4) Were any student predictions on the chalkboard from Phase 1 close to scientifically accurate? Plan to honour those students publicly at the start of Lesson 2 by saying: 'Remember Amina predicted something about air pushing — let us look at whether the evidence supports that.' (5) Did every student successfully draw a Day 1 model? If 20% or more had blank or arrow-free models, begin Lesson 2 with a guided whole-class model-drawing of the jerrycan before moving into new content.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	A sealed, empty 20-litre plastic jerrycan slowly crumpled and collapsed inward on all sides after a small amount of air was removed from inside it — even though no one was touching it. When the cap was reopened, air rushed back in and the jerrycan partially recovered its shape.
What did I learn?	The air around us exerts a real force — called atmospheric pressure — pressing inward on all surfaces from every direction. When the air pressure inside the jerrycan was reduced, the greater atmospheric pressure outside was no longer balanced, and the net inward force crushed the jerrycan. Pressure is force acting per unit area.
How does this explain the phenomenon?	The jerrycan crushed itself because removing air from inside created a pressure difference: the atmospheric pressure pushing inward from outside was greater than the reduced air pressure pushing outward from inside. The unbalanced net force pushed the flexible plastic walls inward. The same air pressure acts on our bodies, but our bodies are filled with fluids and tissues that push back equally — so we do not crush.

LESSON 2 (80 minutes): What Is Pressure? Measuring the Invisible Push — Shoes, Scales, and Force Per Unit Area	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To establish a precise, measurable definition of pressure (P = F/A) by anchoring it to the crushed jerrycan phenomenon and verifying it through direct measurement using students' own shoes, bathroom scales, and graph-paper tracings.
Knowledge	Students will be able to: (a) define pressure as force per unit area (P = F/A); (b) state and use SI units of pressure (Pascals, Pa); (c) explain why the same force produces different pressures depending on contact area; (d) describe atmospheric pressure as the weight of the air column above a surface acting on every exposed face of an object.
Skills	Students will be able to: (a) measure body weight using a bathroom scale and convert Newtons; (b) trace shoe soles on graph paper and estimate contact area in cm^2 and m^2; (c) calculate pressure in Pascals and compare results across different shoe types; (d) construct and interpret a bar chart of pressure values; (e) sketch labelled force-direction arrows on a diagram of the crushed jerrycan.
Attitudes	Students will: (a) appreciate that invisible forces are real, measurable, and governed by mathematical relationships; (b) demonstrate curiosity by connecting everyday objects (shoes, jerrycans, tyres) to a formal physics concept; (c) show responsibility and precision in recording and sharing measurement data.
Key Inquiry Question	How is pressure applied in day-to-day life?
Purpose in Storyline	Lesson 1 left students with a mystery: something invisible crushed the jerrycan. Lesson 2 gives students the first conceptual tool — the definition and formula for pressure — to begin explaining WHAT that invisible push is and HOW it can be quantified. By measuring pressure under their own shoes, students prove that force spread over area is a real, calculable quantity, directly preparing them to ask: 'If my shoe exerts ~20,000 Pa, how much pressure does the entire atmosphere exert on the jerrycan walls?'
Safety Notes	Ensure bathroom scales are placed on a flat, non-slip floor surface. Students should step on and off scales carefully to avoid falls. Graph paper tracings use pencils only — no sharp implements near feet. If hot water was used in Lesson 1 to demonstrate the jerrycan, confirm the jerrycan is at room temperature before any student handles it. Remind students not to place the jerrycan near their faces during any re-demonstration.

B. LESSON OVERVIEW

Students begin by revisiting the crushed jerrycan and posting their first 'What do we think?' and 'What are we wondering?' cards on the Driving Question Board (DQB). The teacher then introduces the formal definition of pressure (P = F/A) through a structured Predict–Observe–Explain sequence using two contrasting objects: a flat book and a sharp pencil pressed onto a palm. Students then conduct a hands-on shoe-pressure investigation — tracing sole outlines on graph paper, reading body weight from bathroom scales, and computing pressure in Pascals. Results are compiled into a class bar chart comparing shoe types (flat sandals, school shoes, sports trainers, heels). Lesson closes with students updating their DQB cards and making a quantitative estimate of atmospheric pressure as a bridge to Lesson 3.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Students are shown the crushed	1. Place the collapsed jerrycan on	Silent Force-Diagramming:	Observable indicator: Do students

📹 **VIDEO:** Pressure and Pascal's principle (part 2) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	jerrycan from Lesson 1 (still collapsed). The teacher places it on the demonstration bench without comment for 60 seconds while students write silently in their notebooks: (a) 'List every force you think is acting ON the jerrycan right now — include direction'; (b) 'Why do you think the outside of the jerrycan won the battle against the plastic?' Students then share with a shoulder partner for 90 seconds. Next, the teacher holds up two objects: a flat 300g hardcover textbook and a sharpened pencil of the same mass. Teacher asks: 'If I press both of these against my palm with exactly the same force, will they feel the same? Why or why not?' Students write a silent prediction (2 sentences) before any discussion occurs.	the bench. Say nothing. Wait 60 seconds — allow cognitive dissonance to re-activate. Then say: 'You have 2 minutes — in your notebook, draw the jerrycan and use arrows to show every force acting on it. Label each arrow.' (WAIT TIME: 2 min silent work). 2. Cold-call 3 students to sketch their force diagrams on the board simultaneously. Ask the class: 'What is the SAME in all three diagrams? What is DIFFERENT?' (WAIT TIME: 15 seconds before taking hands). 3. Hold up the textbook and pencil. Say: 'Same mass — 300 grams each. I press both on my palm. Predict: same sensation or different? Write two sentences. You have 90 seconds — GO.' Do NOT demonstrate yet. Collect 2 written predictions from different corners of the room and read aloud without naming the authors.	Students individually externalise their mental model of forces on the jerrycan before instruction. This surfaces prior conceptions (e.g., 'nothing is pushing it' or 'the air pushed in') and gives the teacher diagnostic data. Sharing diagrams on the board creates productive conflict when students notice they drew arrows in different directions — establishing that the class needs a precise vocabulary to describe this invisible push.	draw force arrows pointing INWARD on all faces of the jerrycan, or only on the top? Students who draw arrows only on top reveal a misconception about pressure acting in all directions — flag these students for targeted questioning during the Explain phase. Collect 4–5 notebooks briefly to scan arrow directions before proceeding.
Observe Phase 📹 **VIDEO:** Pressure and Pascal's principle (part 2) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART A — Pencil vs. Book Demonstration (whole class, 8 min): Teacher presses the flat textbook onto palm, then the pencil tip (safely, moderate force). Students observe and record: sensation described in words, which caused more 'push' sensation, and whether the force applied was the same. PART B — Shoe Pressure Investigation (paired lab, 30 min): Each pair receives: 1 bathroom scale, 1 sheet of graph paper (1 cm² grid), 1 pencil, 1 calculator. Procedure — (i) Student A stands on the scale; partner reads and records weight in kg, converts to Newtons (W = mg, g = 10 N/kg). (ii) Student A steps onto graph paper with ONE shoe; partner traces the outline of the shoe sole carefully.	1. PENCIL vs. BOOK: Press the book firmly on your palm — hold for 5 seconds. Say: 'Observe carefully — not just what you see, but what you predict I feel.' Then press the pencil tip (carefully). Ask: 'What changed? I used roughly the same force both times. Write your observation — 3 words minimum.' Cold-call 2 students. (WAIT TIME: 10 seconds). Say: 'The force was almost the same — so why the different sensation? Hold that question — we'll return to it in 8 minutes.' 2. SHOE LAB SETUP: Distribute materials. Say: 'You have exactly 30 minutes. Your goal is ONE number — the pressure in Pascals under your shoe. Every step of your calculation must be written down — I will check your working, not	Structured Data Collection + Class Data Wall: By pooling measurements from all pairs, students see real variation in pressure values (flat sandals ~15,000–18,000 Pa; school shoes ~20,000–25,000 Pa; pointed heels potentially >100,000 Pa if represented). This variation makes the formula concrete — it is not abstract algebra but a tool that EXPLAINS the variation they measured themselves. The data wall mimics scientific data aggregation and prepares students to reason about atmospheric pressure as a number in the same unit system.	Observable indicators: (a) Can the pair correctly convert kg → N (multiply by 10)? (b) Do they divide body weight by 2 for one foot? (c) Do they convert cm² to m² before computing Pa? Listen for the cm²/m² conversion error — this is the most common mistake. If more than 30% of pairs make this error, pause the class and address it whole-group before the data wall step. Collect one notebook per table group to verify unit conversion working.

	(iii) Student A steps off; pair counts the number of complete 1 cm^2 squares inside the tracing to estimate contact area. Partial squares: count those more than half inside. (iv) Pair calculates pressure: P = F/A, where F = half body weight (one foot) in Newtons, A = area in m^2 (divide cm^2 by 10,000). (v) Pair records shoe type (sandal/school shoe/trainer/other). PART C — Class Data Wall (5 min): Each pair writes their shoe type, area (cm^2), weight (N), and pressure (Pa) on a sticky note and posts it on a shared class data wall organised by shoe type.	just your answer.' Circulate every 4–5 minutes. At each group, ask one probing question — e.g., 'Why are you dividing by 2?' or 'Your area is in cm^2 — is your answer in Pascals yet?' (WAIT TIME: 10 seconds per group before helping). 3. DATA WALL: After posting, stand at the wall and ask: 'Without calculating anything — just LOOKING at this data — what pattern do you notice about shoe type and pressure?' Cold-call 3 students from different parts of the room. (WAIT TIME: 15 seconds).		
Explain Phase 📹 **VIDEO:** [Pressure and Pascal's principle (part 2)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher formally introduces the definition: Pressure = Force ÷ Area (P = F/A), unit = Pascal (Pa = N/m^2). Students complete a structured note frame (provided on ARES platform or as a handout): (a) Write the formula and units; (b) Explain in ONE sentence why the pencil hurt more than the book; (c) Label the three quantities (P, F, A) on a diagram of a shoe on the floor. Students then work in groups of 3 to apply the concept to two Kenya-relevant extension problems: PROBLEM 1 — 'A maize bag weighing 90 kg is placed on a wooden pallet. The pallet has 4 legs, each with a base area of 25 cm^2. What pressure does each leg exert on the warehouse floor in Eldoret?' PROBLEM 2 — 'A Maasai elder uses a traditional rungu (wooden club) with a handle tip area of 2 cm^2. He leans on it with a force of 150 N. A nurse at Kenyatta National Hospital uses a flat crutch base of area 50 cm^2 bearing the same 150 N. Compare the pressures and explain which is	1. FORMAL DEFINITION: Write on board: P = F/A. Say: 'This is the most important equation in today's lesson. But I want YOU to tell ME what it means for the jerrycan.' Point to the collapsed jerrycan. Cold-call 1 student: 'Use the formula — what is F, and what is A, when we talk about the air outside the jerrycan pressing on its walls?' (WAIT TIME: 15 seconds). 2. NOTE FRAME: Give 4 minutes for silent written completion of the note frame. Then pair-share for 90 seconds. Cold-call one pair to read their one-sentence pencil explanation aloud. 3. EXTENSION PROBLEMS: Assign groups. Say: 'You have 10 minutes. Show ALL unit conversions. If your units don't cancel to give Pa (N/m^2), your answer is wrong — check your working.' Circulate. At the whiteboard presentation, ask: 'Which answer surprises you most — and why does it connect back to the jerrycan?' (WAIT TIME: 10 seconds before taking responses). 4. JERRYCAN LINK: Say: 'We now know pressure is F/A. The	Anchored Application Problems: By embedding P = F/A in recognisable Kenyan contexts (maize warehouse in Eldoret, Maasai rungu, KNH crutch), students see the formula as a tool for reasoning about the real world rather than an abstract symbol. Presenting solutions on mini-whiteboards allows rapid whole-class formative assessment and creates a norm of mathematical communication. The closing jerrycan link ensures every new piece of knowledge is explicitly tethered to the anchoring phenomenon.	Observable indicators: (a) Students correctly identify F (body weight in N) and A (shoe area in m^2) when labelling the diagram. (b) In Problem 1, students divide total force by 4 legs before dividing by leg area. (c) In Problem 2, students correctly conclude the rungu exerts higher pressure and connect this to potential floor damage OR to the jerrycan walls — any valid pressure-direction reasoning is acceptable. Flag groups that invert the formula (A/F) for immediate reteaching.

	safer for a soft floor.' Groups present solutions on mini-whiteboards.	atmosphere is pushing on the jerrycan with some force over its entire surface area. In Lesson 3 we will calculate exactly how big that number is. Make a prediction now — write it in your notebook: is atmospheric pressure bigger or smaller than the pressure under your shoe?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Pressure and Pascal's principle (part 2) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes of different colours. On COLOUR 1 (e.g., yellow): 'What do I now THINK is causing the jerrycan to crush?' — students write their current best explanation using the word 'pressure' and 'force per unit area'. On COLOUR 2 (e.g., blue): 'What am I still WONDERING?' — students write one specific question they cannot yet answer (e.g., 'How do we know how strong atmospheric pressure actually is?' or 'Does pressure in water work the same way as pressure in air?'). Students post both cards on the class DQB. Teacher reads 5–6 blue 'wondering' cards aloud and groups them into clusters on the board (e.g., 'Questions about HOW BIG atmospheric pressure is', 'Questions about pressure in water', 'Questions about why our bodies don't get crushed'). Teacher labels each cluster with a sticky arrow and says: 'These clusters are our roadmap. By Lesson 9, every one of these clusters will be answered.'	1. Distribute sticky notes. Say: 'You have 3 minutes — silent writing only. Your yellow card must contain the phrase force per unit area. Your blue card must be a genuine question you cannot answer right now — not a question you already know the answer to.' (WAIT TIME: 3 min silent). 2. Students post cards. Teacher reads 5–6 blue cards aloud (without naming authors). Ask class: 'These two blue cards seem related — can anyone tell me WHY?' Cold-call 2 students. (WAIT TIME: 10 seconds). 3. Cluster the cards visibly. Say: 'Notice that nobody has posted a card asking what pressure means anymore — because we answered that today. This is how science works: we eliminate questions as we gather evidence, and new questions appear.' 4. Point to the jerrycan. Say: 'By Lesson 3, the yellow cards will get more detailed. Keep your notebooks — we'll look back at your Lesson 1 predictions and your Lesson 2 predictions side by side.'	Two-Colour DQB Protocol: Separating 'what I think' (yellow) from 'what I wonder' (blue) trains students to distinguish between claims and questions — a core science and engineering practice. Clustering the blue cards gives students ownership of the lesson sequence and demonstrates that their questions are driving the investigation, not a predetermined textbook chapter. This reinforces the storyline coherence of the 9-lesson unit.	Observable indicators: (a) Yellow cards use 'pressure', 'force', and 'area' correctly — not vague phrases like 'the air did it'. (b) Blue cards are specific and answerable by science (not philosophical). (c) Students who cannot write a yellow card using the formula likely need re-engagement with the Explain phase before Lesson 3. Collect yellow cards from 5 students who struggled during the lab for review before Lesson 3.
Model Building Phase 📹 **VIDEO:** Pressure and Pascal's principle (part 2)	Students return to their Lesson 1 force diagram of the jerrycan (or draw a fresh one if Lesson 1 was introductory). Using what they now know about pressure (P = F/A), they REVISE their model by: (a) Adding labelled arrows on ALL six	1. Say: 'Open your notebook to your Lesson 1 jerrycan diagram. You have 6 minutes to revise it using everything from today. You MUST add arrows on all six faces — not just the top. You MUST write the sentence frame below the	Iterative Model Revision with Green-Pen Tracking: Asking students to circle changes in green makes the evolution of their thinking visible to themselves and to the teacher. This metacognitive move reinforces that scientific	Observable indicators: (a) Revised diagrams show inward arrows on ALL faces (not just top/sides). (b) Sentence frame correctly identifies balanced pressure as the reason the full jerrycan stayed intact, and pressure imbalance (less inside)

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	faces of the jerrycan (top, bottom, four sides) pointing INWARD — representing atmospheric pressure; (b) Adding a second set of arrows pointing OUTWARD on the INSIDE — representing air pressure from inside pushing out; (c) Writing one sentence under the diagram: 'Before the air was removed, the jerrycan did not collapse because ___. After air was removed, the jerrycan collapsed because ___.' Students compare their revised model with their Lesson 1 model and circle what has changed in GREEN pen. In groups of 4, students share one 'biggest change' in their model and explain why their thinking changed.	diagram.' (WAIT TIME: 6 min silent revision). 2. Circulate. Look specifically for students who only draw arrows on the top face — gently prompt: 'Is the air only above the jerrycan, or is it all around it?' (WAIT TIME: 10 seconds for student response before guiding further). 3. Group share: 'Tell your group the ONE thing in your model that changed the most from Lesson 1 to Lesson 2. Use the words force, area, or pressure in your explanation.' Cold-call 2 groups to share with the class. (WAIT TIME: 10 seconds). 4. Close by pointing to the DQB. Say: 'Your model today answers PART of the driving question. Next lesson, we will find out exactly HOW BIG that invisible push is — and why it doesn't crush your body the way it crushed the jerrycan.'	models are not fixed answers but living representations that improve with evidence. Requiring arrows on all six faces directly targets the common misconception that atmospheric pressure acts only downward — a misconception that will resurface in the atmospheric pressure lesson (Lesson 3).	as the reason it collapsed. (c) Students who still draw arrows only from above, or who write 'the air got heavier' in their sentence frame, are identified for Lesson 3 small-group support. Photograph 4–5 revised diagrams using the ARES device camera for the teacher reflection record.

D. TEACHER REFLECTION

After the lesson, review the photographed model diagrams and the yellow DQB cards to answer: (1) What percentage of students correctly placed inward pressure arrows on all six faces of the jerrycan? If fewer than 70% did so, open Lesson 3 with a whole-class arrow-placement activity before introducing atmospheric pressure values. (2) Did students successfully convert cm^2 to m^2 in the shoe lab? If the conversion error was widespread, prepare a brief unit-conversion warm-up for Lesson 3. (3) Read the blue 'wondering' cards: do students' questions cluster around 'how big is atmospheric pressure?' — this is the ideal setup for Lesson 3. If questions cluster instead around 'pressure in water', consider briefly acknowledging that thread and signposting it to Lessons 4–5. (4) Were any students reluctant to stand on the scale in front of peers? If so, offer the alternative of measuring the weight of a filled water bottle (known volume) in future lessons, or allow students to use a classmate's weight data anonymously.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the collapsed jerrycan (phenomenon revisit), felt the difference between a flat book and a pencil tip pressed with equal force, and collected shoe-sole area and body-weight data to calculate pressure in Pascals across different shoe types.
What did I learn?	Pressure is defined as force per unit area ($P = F/A$) and is measured in Pascals ($Pa = N/m^2$). The same force produces higher pressure when applied over a smaller area. Atmospheric pressure acts inward on all faces of the jerrycan; when the inside air pressure was reduced, the forces became unbalanced and the jerrycan collapsed.
How does this explain the	Students explained — using labelled force diagrams and the sentence frame — that the full jerrycan remained intact because

phenomenon?	inside and outside air pressure were balanced (net force = zero on each face), but removing air from inside reduced the outward push, leaving the larger atmospheric pressure outside to crush the walls inward. This explanation is partial: Lesson 3 will quantify exactly how large atmospheric pressure is and why our bodies are not similarly crushed.

LESSON 3 (40 minutes): Going Deeper: Factors Affecting Pressure in Liquids	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Students investigate and identify the three factors that determine pressure in a liquid — depth, density, and gravitational field strength — building the conceptual foundation for deriving P = rho x g x h in Lesson 4.
Knowledge	Students will be able to state that pressure in a liquid increases with depth, increases with the density of the liquid, and depends on gravitational field strength; and identify these as the three factors that determine liquid pressure.
Skills	Students will design and conduct a fair experiment using modified plastic bottles and water to measure how liquid pressure changes with depth; record data in organised tables; and plot and interpret a graph of pressure versus depth.
Attitudes	Students will demonstrate curiosity and persistence when results are unexpected, collaborate equitably in groups, and show care in handling water and glassware to avoid spills.
Key Inquiry Question	How does pressure in fluids affect the design of machines and structures?
Purpose in Storyline	The jerrycan collapsed because unbalanced pressure acted on it from outside. But where does liquid pressure come from, and what controls how strong it is? This lesson moves students from atmospheric pressure (Lesson 2) into liquid pressure, generating the evidence base they need to unlock the quantitative equation in Lesson 4 and to later explain dam walls, water towers, and deep-sea diving in Kenyan contexts.
Safety Notes	Warn students that cut plastic bottles have sharp edges — use masking tape on cut rims. Ensure water spills are mopped immediately to prevent slips. Do not use hot water in this experiment. Supervise the use of pins or nails used to make holes in bottles.

B. LESSON OVERVIEW
Students begin by reconnecting to the crushed jerrycan and posing the question of whether liquid pressure behaves like atmospheric pressure. They predict what will happen to water jets from holes at different depths in a plastic bottle, then conduct the experiment to observe that deeper holes produce faster, farther jets. A second investigation compares water and a dense salt solution to surface the role of density. Students organise their data, identify the three factors controlling liquid pressure, and update their Driving Question Board with new evidence. They close by sketching a particle-level model explaining why deeper liquid means more pressure.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos	Students sit in groups of four. The teacher holds up a 1.5-litre transparent plastic bottle that has three small holes punched at different heights — near the top, at mid-height, and near the base — each sealed with a small piece of	Teacher holds the bottle at eye level and says: 'Before I remove these tapes, I want every person to commit to a prediction in writing — do not look at your neighbour yet.' Wait 60 seconds in silence. Then: 'Now compare with your	Individual written prediction before group discussion prevents anchoring bias and ensures every student commits to a reasoning chain. Recording group predictions publicly creates accountability and motivates careful observation in	Teacher reads the half-sheet predictions collected from groups. Look for whether students reference force, area, or the weight of water above a point — these show transfer from Lesson 1 and 2 learning. Students who

📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	masking tape. Students are told the bottle will be filled with water and all three tapes removed simultaneously. Each student individually writes in their notebook: (1) which hole will produce the strongest jet, (2) how far from the bottle each jet will land, and (3) a one-sentence reason connecting their prediction to what they learned about atmospheric pressure and the jerrycan in Lessons 1 and 2. Students then share predictions within their group of four and agree on a group prediction to record on a half-sheet of paper.	group. What is your group betting on — and why?' Circulate and listen for misconceptions such as all jets being equal or the top hole being strongest because it is closest to open air. After 90 seconds of group discussion, cold-call two groups with contrasting predictions and ask: 'What evidence from our jerrycan lessons supports your reasoning?' Do NOT correct predictions — affirm that all ideas will be tested. Write the two competing class predictions on the board under the heading 'Our Bets'.	the next phase.	predict all jets are equal or that the top jet is strongest will need targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** [Pascal's Triangle - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives: one 1.5-litre transparent bottle pre-drilled with three holes at heights approximately 5 cm, 12 cm, and 20 cm from the base, sealed with masking tape; a ruler; a large basin or tray; a jug of water; and a data recording sheet. Groups fill the bottle over the basin, then simultaneously remove all three tapes and observe the jets. They use the ruler to measure the horizontal distance each jet travels before hitting the tray surface, and record results in a three-column table (hole height from base, horizontal distance of jet, relative strength ranking). Groups repeat the fill-and-release twice more for consistency and average their results. In a second five-minute investigation, groups are given a second identical bottle but filled with a dense salt solution (prepared by the teacher: 200 g of table salt dissolved in 1 litre of water, coloured with food dye for visibility). They compare the jet distances from the same depth holes and record observations.	Before groups begin: 'You are scientists right now. Your job is to observe exactly what happens and measure it — do not explain yet, just record what you see.' Emphasise fair testing: 'What must you keep the same each time you fill the bottle so your results are fair to compare?' Wait 30 seconds for student responses — draw out: same water level when tapes are removed, same bottle orientation, same measurement method. Circulate during the experiment. When a group notices the bottom jet travels farthest, resist confirming — instead ask: 'Interesting — what do you think is causing that difference? Write it in your notes — we will come back to it.' For the salt-solution bottle, prompt: 'The salt water jet from the same depth — is it different from the fresh water jet? What does that suggest to you?' If time is short, the salt-solution comparison can be demonstrated by the teacher at the front rather than done by every group.	The multi-hole bottle is a classic discrepant event that makes liquid pressure differences visible and measurable. Repeating the measurement three times and averaging introduces scientific rigour appropriate for Grade 10 CBE. The salt-solution comparison surfaces the density variable without the teacher naming it first, preserving student agency in pattern recognition.	Teacher checks each group's data table during circulation. Key diagnostic: are students recording the hole height from the BASE (not from the top)? Correct measurement orientation matters for the graph in the Explain phase. Also check that students record both the fresh-water and salt-water results in clearly labelled separate columns — this distinction is needed for the density discussion.

Explain Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher leads a class discussion to draw out patterns from the data. Students share their average jet-distance measurements; the teacher records them in a class data table on the board with columns for hole height, average jet distance (fresh water), and average jet distance (salt water). Students are then asked to plot a graph of jet distance (proxy for pressure) on the y-axis versus hole height from the base (depth of water above the hole) on the x-axis using graph paper or their notebooks. They describe the shape of the graph in one sentence. Next, the teacher introduces the three-factor framework: the class identifies from their data that pressure increases with depth (deeper hole, longer jet), pressure increases with density (salt-water jet travels farther than fresh-water jet from same depth), and the teacher introduces gravitational field strength (g = 10 N/kg) as the third factor by posing a thought question: 'If you took this same bottle of water to the Moon where gravity is much weaker, would the jet travel as far? Discuss.' After 60 seconds of think-pair-share, students record the three factors in a structured summary: Factor 1 — Depth (h); Factor 2 — Density (rho); Factor 3 — Gravitational field strength (g). Students write the summary statement: 'Liquid pressure depends on the depth of the liquid, the density of the liquid, and the strength of gravity at that location.' Finally, the teacher connects back to the jerrycan: 'The water inside our jerrycan was also exerting pressure on the walls — and now we can say that pressure	When recording class data: 'I notice Group 3 has a slightly different value for the middle hole — what might explain that variation?' Accept answers about measurement error, water level not being exactly the same, or the hole size varying. This normalises scientific uncertainty. When introducing the Moon question: 'I am going to give you 60 seconds of silence to think about this before you talk — use it.' WAIT 60 seconds. After pairs share, say: 'So gravity matters. We cannot change gravity easily in the lab, but we have changed depth and density today — and both changed the pressure. Three factors, all revealed by one plastic bottle.' When presenting the three-factor summary, ask students to say the three factors aloud together as a class once, then write them individually. Use a Kenyan context to anchor each factor: Depth — 'Think of a diver in the Indian Ocean off Mombasa going deeper and deeper'; Density — 'Think of Lake Magadi, which is so dense with soda ash that objects float more easily — its pressure at a given depth is higher than fresh water'; Gravity — 'On a future Kenyan space mission, a water system would need redesign because g is different.'	Graphing jet distance versus depth transforms a qualitative observation into a quantitative pattern and builds the graphing skill needed for Boyle's Law in Lesson 7. The Moon thought experiment isolates the gravity variable without requiring a second experiment. The three-factor structured summary gives students a reusable scaffold they will use in Lesson 4 to derive P = rho x g x h.	Collect graphs from each student. Check that: the graph is labelled with axes and units, the trend is correctly described as increasing (not decreasing or flat), and the student can articulate why jet distance represents pressure. Use exit observation: ask three students at random to state the three factors without looking at their notes.

	depended on the depth of water and its density. When we poured the water out, we removed that outward push too. So the jerrycan had to deal with atmosphere pushing in from outside, with much less pushing back from inside.'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Pascal's Triangle - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each student receives two sticky notes (or small paper squares if sticky notes are unavailable). On the first note they write one new thing they now KNOW about liquid pressure from today's lesson. On the second note they write one new question that today's investigation has raised for them — especially questions about the jerrycan, dams, water towers, or the ocean. Students post their notes on the class Driving Question Board under the headings 'New Evidence We Have' and 'New Questions We Are Asking'. The teacher then reads aloud three or four of the new questions and invites the class to vote (show of hands) on which question they most want to answer next. The teacher draws a box around the top vote-getter and writes beside it: 'Watch for this in Lesson 4.'	Say: 'You have spent this lesson gathering evidence. Evidence is only useful if we connect it to our big question. Take two minutes — no talking — to write your two notes.' WAIT 2 minutes. As students post notes, scan for questions that anticipate P = rho x g x h or ask about dams and water towers — these validate the storyline sequence. If no student raises a question about calculating pressure numerically, the teacher adds a teacher card: 'We know the three factors — but can we calculate exactly how much pressure acts at a specific depth? How?' This seeds Lesson 4. Avoid answering any questions now. Say: 'These questions are our engine. We will come back to every single one before Lesson 9.'	The DQB ritual maintains the coherent investigative narrative across all nine lessons and gives students agency over the direction of inquiry. Voting on which question to pursue first gives students a sense of ownership over the curriculum sequence.	The 'New Evidence' notes function as a quick knowledge check. Teacher collects and reads all notes before Lesson 4 to identify students who have not yet distinguished between depth and density as separate factors, or who believe pressure in liquids acts only downward (a common misconception to address at the start of Lesson 4).
Model Building Phase 📹 **VIDEO:** [Pascal's Triangle - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12	Students return to the cumulative model they began in Lesson 1 (a sketch of the jerrycan with force arrows). They are asked to add a new section to their model labelled 'Liquid Pressure'. In this section they draw: (1) a tall column of water (like a water tower or dam) with horizontal arrows pointing outward from the column at three heights — a short arrow near the top, a medium arrow at mid-height, and a long arrow at the base — to show that outward pressure	Circulate and look at model sketches. Prompt students who draw arrows only pointing downward: 'Remember Lesson 2 — the atmosphere pushed in all directions on the jerrycan. Does liquid pressure also act in all directions, or only downward? What does our bottle experiment suggest?' (The jets came out horizontally, demonstrating that liquid pressure acts outward, not just downward.) For the particle explanation, prompt: 'Imagine you	The cumulative model is the primary evidence-gathering artefact of the nine-lesson unit. Adding to it each lesson builds scientific modelling skills and creates a visual record of conceptual development. The particle-level explanation bridges macroscopic observation (jet distance) to microscopic mechanism, which is a key KICD competency for Grade 10 Physics.	Teacher does a gallery walk of five to six models before the end of class. Look for: correct direction of pressure arrows (outward/sideways, not only downward), increasing arrow length with depth, presence of all three factor labels, and a particle-level explanation in the student's own words. Students whose models show only downward arrows are flagged for a brief clarifying conversation at the start of Lesson 4.

🔍 Search ARES for similar readings	increases with depth; (2) labels on the arrows showing the three factors (h, rho, g); (3) a brief particle explanation in their own words: 'At greater depth there are more water particles above pressing down, so the pressure is higher.' Students who finish early are challenged to add a second sketch showing what happens to the arrows if fresh water is replaced with salt water at the same depths.	are a tiny particle at the bottom of the bottle. How many other water particles are sitting above you compared to a particle near the top? What does that do to the push you feel?' WAIT 30 seconds after posing this question before accepting answers. Close the model-building phase by saying: 'Your model now shows that pressure in liquids is not flat — it has a shape. Next lesson we will give that shape an equation.'		

D. TEACHER REFLECTION

After the lesson, consider: Did all groups successfully measure jet distances, or did some groups struggle with the bottle setup? If setup issues occurred, prepare a teacher-demonstrated version as a backup for any future lesson using this investigation. Did students spontaneously mention density as a factor before the salt-water comparison, or did it only emerge from the second investigation? If students raised it early, they may be ready to move faster into P = rho x g x h in Lesson 4. Were there any student questions on the DQB that go beyond the scope of this unit — for example, about surface tension or capillary action? Note these for differentiation. Most importantly: do students now hold a clear three-factor mental model of liquid pressure, or are they conflating depth and the volume of water? This distinction is critical before deriving the equation in Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Water jets from holes at different depths in a plastic bottle: the deeper the hole, the farther the jet traveled. The jet from a salt-water bottle at the same depth traveled farther than the jet from a fresh-water bottle.
What did I learn?	Pressure in a liquid depends on three factors: depth (h), density of the liquid (rho), and gravitational field strength (g). Pressure increases as depth increases and as density increases.
How does this explain the phenomenon?	At greater depth in a liquid, there is more liquid above that point pressing down, so the pressure is higher. Denser liquids have more mass per unit volume, so they exert more pressure at the same depth. This explains why water at the base of a dam wall pushes much harder on the wall than water near the surface, and it begins to explain why the jerrycan walls experienced large inward forces when the inside air was removed.

LESSON 4 (80 minutes): The Invisible Giant: Proving the Atmosphere Has Weight and Pushes in All Directions	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To gather direct experimental and historical evidence that the atmosphere exerts pressure in all directions, and to use this evidence to explain why the jerrycan collapsed when air was removed from inside.
Knowledge	The learner describes atmospheric pressure as the weight of the air column above a surface acting in all directions; explains that removing air from inside a container creates a pressure imbalance that causes collapse; and identifies the mercury barometer as a device that measures atmospheric pressure.
Skills	The learner conducts a collapsing-can experiment, records qualitative observations, interprets a mercury barometer diagram, and revises a scientific model based on new evidence.
Attitudes	The learner appreciates that invisible forces govern everyday phenomena and demonstrates curiosity and open-mindedness when experimental results contradict prior assumptions.
Key Inquiry Question	How is pressure applied in day-to-day life? / How does pressure in fluids affect the design of machines and structures?
Purpose in Storyline	Lesson 4 is the critical 'aha' lesson in the evidence-gathering sequence. Students have already measured pressure on solids and explored liquid pressure. Now they gather direct evidence — through a live classroom experiment, a historical video demonstration, and a barometer diagram — that the atmosphere itself is a fluid that presses with enormous force in every direction. This is the missing piece that fully explains the jerrycan: the plastic walls did not collapse because they were weak; they collapsed because, once internal air was removed, the colossal and ever-present outward push of the atmosphere lost its inward partner — and the atmosphere won. Students revise their Driving Question Board cards to reflect this causal understanding.
Safety Notes	SAFETY: The collapsing-can demonstration uses boiling water and a heat source (jiko/gas burner). Only the teacher or a designated adult should handle the hot tin and boiling water. Students must remain seated at least 1.5 metres from the heat source. Use metal tongs — never bare hands — to invert the hot tin into cold water. Warn students the implosion produces a sudden loud bang; students with sensitivity to sudden sounds should be forewarned. Ensure the workspace is clear of flammable materials. If a full gas burner is unavailable, hot water from a thermos flask poured into the tin for 60 seconds is a safe alternative heat source.

B. LESSON OVERVIEW

Students open the lesson by revisiting their jerrycan collapse predictions from earlier lessons, then make a new focused prediction: what will happen to a sealed metal tin after boiling water is poured out and the tin is immediately sealed and cooled? The teacher conducts the collapsing-tin demonstration live. Students observe, record, and compare with the jerrycan event. They then watch a short video clip of the Magdeburg Hemisphere demonstration and study a labelled mercury barometer diagram to triangulate evidence that atmospheric pressure is real, massive, and omnidirectional. Through a structured 'Evidence–Reasoning–Claim' sensemaking routine, students construct the causal explanation for the jerrycan collapse. The lesson closes with a DQB card revision session and a 3-2-1 exit ticket.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students are shown a clean, empty 500 mL metal tin (e.g., a cooking-fat or paint tin) with its lid and a basin of cold water placed at the front. The teacher says: 'I am going to put a small amount of water in this tin, heat it until steam pours out of the open top, then quickly seal the lid and place the tin in this cold water basin. Before I do anything — write in your notebooks exactly what you predict will happen to the tin, and WHY. You have 3 minutes. Be specific.' Students write individual predictions silently. They then share with a partner (Think-Pair-Share) and pairs record one joint prediction on a sticky note placed on the class Prediction Wall under the heading 'Tin Experiment — Our Best Prediction.' The teacher also directs students to re-read their original jerrycan prediction cards from Lesson 1 and asks them to flag any part of their earlier reasoning they now want to change.	1. Place the tin, basin of cold water, and heat source visibly at the front BEFORE the lesson begins — do not explain them yet; allow curiosity to build as students enter. 2. After posing the prediction task, say exactly: 'Write your own idea first — do not look at your neighbour's book yet. Your prediction is not marked right or wrong; I want your honest thinking.' Set a visible 3-minute timer. 3. After individual writing, say: 'Now turn to your partner. You have 90 seconds — share your prediction and agree on one joint statement.' 4. Cold-call 3 pairs (choose one high-confidence pair, one uncertain pair, one pair whose sticky note shows a misconception). Ask each: 'Tell us your prediction AND your reasoning — what makes you think that?' Apply 12-second WAIT TIME after each pair speaks before responding. 5. Record the three predictions on the board in a T-chart: PREDICTION | REASONING. Do NOT confirm or deny any prediction. Say: 'Keep these in mind. Evidence will answer for us.'	STRATEGY — Think-Pair-Share with Prediction Wall: This activates prior knowledge and surfaces existing mental models about air and containers. Recording predictions publicly creates accountability and cognitive investment — students will be motivated to reconcile their prediction with observation. Flagging earlier jerrycan cards builds metacognitive awareness of how scientific thinking evolves with evidence.	INDICATOR: Teacher scans sticky notes and notebook entries for the quality of causal reasoning. Target indicator — does the student mention air, pressure, or force in their reasoning, or do they attribute the prediction purely to heat/temperature? Students who make no mention of air or pressure are flagged for targeted questioning during the Observe phase. Teacher records initials of 3–4 such students to cold-call specifically in Phase 2.
Observe Phase 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook)	ACTIVITY A — Live Collapsing Tin Demonstration (12 min): The teacher conducts the demonstration step by step, narrating each action. Students complete a structured observation table in their notebooks with columns: WHAT I SEE | WHAT I HEAR | WHAT I FEEL (from a distance) | QUESTIONS THIS RAISES. The tin implodes dramatically when placed in cold water. Students record all sensory observations immediately. ACTIVITY B — Magdeburg	ACTIVITY A: 1. Narrate each step slowly and clearly: 'I am adding about 50 mL of water to the tin… I am placing it on the heat source… watch the steam rising from the open top — that steam is PUSHING air out of the tin… now I am sealing the lid tightly…' Pause 5 seconds. 'Now — into the cold water.' Allow the implosion to happen. After the sound and collapse, say NOTHING for 10 seconds. Let the observation sink in. Then say: 'Write. Everything. You just saw.' Give 2 minutes of	STRATEGY — Multi-Source Evidence Triangulation: Three distinct pieces of evidence (live experiment, historical demonstration, instrument diagram) all point to the same phenomenon — atmospheric pressure is real and enormous. Triangulation builds scientific confidence: if only one experiment showed this, it might be a fluke. When three independent sources agree, the conclusion becomes robust. This mirrors how professional scientists build	INDICATOR 1 (Tin Demo): Students' observation tables should include at least one entry in EACH column. Incomplete tables (e.g., only 'the tin got squashed') indicate surface-level observation — teacher prompts these students: 'What did you HEAR? What direction did the sides move?' INDICATOR 2 (Video): Focus-question responses should name air/atmospheric pressure as the force — not 'the pump' or 'the machine.' Students who write 'the pump is too weak' are still

Source: CK-12 🔍 Search ARES for similar readings	Hemispheres Video (6 min): Students watch a 4-minute video clip showing the 1654 Magdeburg Hemispheres demonstration (Otto von Guericke's experiment where two hollow bronze hemispheres held together by atmospheric pressure could not be pulled apart by teams of horses once the air inside was pumped out). Before the video starts, the teacher gives a focus question written on the board: 'What is STOPPING the horses from pulling the hemispheres apart? Write the answer as you watch.' Students write during the video. ACTIVITY C — Mercury Barometer Diagram (7 min): Students receive a printed or projected labelled diagram of a mercury barometer (Torricelli's barometer). Key labels include: vacuum at the top, mercury column (~760 mm), open mercury trough, atmosphere pressing down on trough. Students annotate the diagram by drawing arrows showing WHERE the atmospheric pressure acts and labelling the pressure value at sea level (101,325 Pa ≈ 101.3 kPa). They answer: 'Why does the mercury NOT fall completely out of the tube?'	silent writing. 2. Cold-call 4 students (include the 3–4 flagged in Phase 1): 'What did you observe? Be precise — use describing words.' WAIT 12 seconds after each. 3. Ask the class: 'How is THIS tin similar to the jerrycan from Lesson 1? How is it different?' Cold-call 2 students. ACTIVITY B: Before video: 'Your only job during this video is to answer one question — write it down: What is stopping the horses?' During video, circulate and glance at notebooks. After video: cold-call 3 students with the focus question. Ask a follow-up: 'If you pumped the air BACK into the hemispheres, what do you predict would happen? Why?' WAIT 10 seconds. ACTIVITY C: Circulate as students annotate. Ask targeted questions to seated students: 'Where exactly is the atmosphere pushing in this diagram?' and 'What would happen to the mercury level if you climbed to the top of Mount Kenya — would it go up or down?' Accept and probe all responses without confirming immediately.	consensus and directly models NGSS Science and Engineering Practice 3 (Planning and Carrying Out Investigations) and Practice 8 (Obtaining, Evaluating, and Communicating Information).	attributing agency to the device rather than the atmosphere. INDICATOR 3 (Barometer): Arrows on the diagram should point DOWNWARD from the atmosphere onto the mercury trough surface. Students who draw arrows pointing upward from the mercury have a persistent misconception about direction of atmospheric pressure that must be addressed in Phase 3.
Explain Phase 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure	Students work in groups of four to complete an 'Evidence–Reasoning–Claim' (ERC) graphic organiser on a large sheet of paper. The organiser has three columns: EVIDENCE (what we observed/saw/read), REASONING (the science principle that connects the evidence to our answer), and CLAIM (our answer to the question). The target question printed at the top of the organiser is: 'Why did the jerrycan	1. Distribute ERC organiser sheets and say: 'Your group's job is to answer the jerrycan question using TODAY's evidence — not your opinions, but the specific things we observed and read today. You have 10 minutes.' Set a timer. Circulate and listen — do not give answers. Ask probing questions: 'Which piece of evidence is strongest? Why?' and 'What is the science PRINCIPLE behind what you observed in the tin?' 2. During	STRATEGY — Evidence-Reasoning-Claim (ERC) Organiser: ERC is a structured argumentation scaffold that mirrors the logic of scientific writing and directly embeds NGSS Practice 7 (Engaging in Argument from Evidence). By requiring students to separate WHAT they saw from WHY it matters, the organiser prevents the common error of restating observations as explanations. The gallery walk	INDICATOR: ERC organisers must show a causal chain — evidence → pressure principle → jerrycan explanation. Incomplete chains (e.g., 'the tin crushed because of steam') without naming pressure imbalance as the mechanism are flagged. During gallery walk, the teacher counts how many posters include the phrase 'pressure imbalance' or 'unequal pressure' — this is the target conceptual language.

(Flexbook) Source: CK-12 🔍 Search ARES for similar readings	crush itself — and what does today's evidence prove about the atmosphere?' Each group must include evidence from ALL THREE activities (tin, video, barometer). Groups then complete a 'Pressure Balance Diagram': a simple before/after sketch of the jerrycan, labelling arrows for atmospheric pressure pushing IN from outside and internal air pressure pushing OUT from inside, and showing what changes when internal air is removed. Groups present their ERC claims to the class in 60-second gallery-walk format (posters are posted on walls; students rotate and add a green tick if they agree with a claim or a yellow question mark if they disagree or need more explanation). The teacher closes this phase by delivering a 3-minute direct explanation of atmospheric pressure: definition, cause (weight of air column above), omnidirectional nature, and value at sea level (~101.3 kPa). Teacher uses the analogy of students swimming in Lake Victoria — the water pushes from all sides — then extends: 'We swim in a sea of air. The air also pushes from all sides. Always.'	gallery walk (4 minutes): circulate and read group posters. Identify 2 posters with strong causal language and 1 poster with a gap (e.g., correct observation but missing the pressure balance concept) for targeted whole-class discussion. 3. After gallery walk, cold-call the group with the clearest pressure-balance explanation: 'Group 3 — read us your CLAIM. Then tell us which piece of evidence you found most convincing and why.' WAIT 12 seconds. 4. Direct explanation (3 min): Use the board to draw the Lake Victoria swimming analogy, then the jerrycan pressure-balance diagram with labelled arrows. Say explicitly: 'Atmospheric pressure at sea level — right here in Mombasa — is approximately 101,300 Pascals. That means the atmosphere is pressing on every square metre of your body with about 101,300 Newtons of force. You do not feel it because the same pressure exists inside your body, pushing outward. The jerrycan had that same balance — until we removed the air inside. Then the atmosphere had nothing to push against. And it won.' 5. Write the key equation on the board: $P_{atm} \approx 101.3$ kPa = 101,300 Pa. Ask the class: 'In what unit is this measured?' Cold-call 2 students.	adds peer evaluation, building critical thinking and collaborative norms. The teacher's Lake Victoria analogy bridges the abstract (omnidirectional fluid pressure) to a felt, embodied Kenyan experience that students can visualise.	TARGET: at least 4 of 6 groups use correct causal language. Groups that do not achieve this are specifically addressed in Phase 3's direct explanation. Post-explanation, teacher poses an exit-check question verbally: 'Thumbs up if you can now explain, in one sentence, why the jerrycan collapsed' — then cold-calls 2 thumbs-up students and 1 hesitant student to verify.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Introduction to experiment design Source: Khan	Each student retrieves their personal DQB card from the class board (index cards posted in Lesson 1 with their original jerrycan questions and predictions). Students now work individually to revise their card in a different coloured pen/marker, adding: (1) one new ANSWER	1. Distribute DQB cards back to students and say: 'Pick up a pen or marker of a DIFFERENT colour from what you used before. You have 5 minutes — add your new answer and your new question. The new colour shows how your thinking has grown since Lesson 1.' 2. After cards are reposted,	STRATEGY — Visible Thinking Revision (Colour-Coded DQB Cards): Using a different pen colour to add new thinking makes conceptual growth literally visible — to the student and to the teacher. This embeds metacognition: students must distinguish what they knew before	INDICATOR: The teacher checks that each revised DQB card contains BOTH a new answer AND a new question. Cards with only a new answer (no new question) indicate a student who thinks the topic is 'done' — a misconception about the nature of scientific inquiry. Cards with only a

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	they can now give confidently based on today's evidence, (2) one new QUESTION that today's lesson opened up (e.g., 'Does atmospheric pressure change with altitude? What happens on Mount Kenya?', 'Why doesn't our body crush like the jerrycan?', 'How did Torricelli know the mercury height was caused by air and not something else?'). Students then post updated cards back on the DQB. The teacher reads 5–6 new questions aloud and groups them into emerging themes on the board (altitude, body pressure, historical measurement, machines). The class votes on which theme they are most curious about — this vote informs the teacher's emphasis in Lesson 5.	read 5–6 new student questions aloud without attribution. For each, ask: 'Who else had a question like this? Raise your hand.' This validates curiosity and shows students their questions are scientifically meaningful. 3. Cluster questions on the board using sticky arrows or drawn circles. Label clusters: e.g., 'ALTITUDE cluster', 'BODY cluster', 'HISTORY cluster', 'MACHINES cluster'. 4. Say: 'In the coming lessons, we will chase each of these clusters with more evidence. For now — which cluster are you MOST curious about?' Conduct a show-of-hands vote. Record the tally visibly. 5. Close by connecting to the driving question: 'We started asking why the jerrycan crushed itself. Today we proved the invisible giant — the atmosphere — was responsible. But the question on our board asks about more than jerrycans: it asks about mountains, saving lives, and powering machines. Those answers are coming.'	from what they know now. The class vote on emerging question clusters transfers ownership of the inquiry to students, sustaining intrinsic motivation and modelling how scientific communities prioritise research questions.	new question and no answer may indicate a student who did not consolidate understanding from Phase 3 and needs follow-up. The teacher collects and photographs all updated DQB cards before the next lesson for formative records.
Model Building Phase 📹 **VIDEO:** Introduction to experiment design Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES	Students open their cumulative model pages (begun in Lesson 1) — a diagram of the jerrycan that they have been revising across the sequence. Today's addition is specific: students must add ATMOSPHERIC PRESSURE ARROWS pressing inward from ALL FOUR SIDES and the TOP of the jerrycan from outside, and INTERNAL AIR PRESSURE ARROWS pushing outward from all sides on the inside. They label the balanced state (jerrycan intact) and the imbalanced state (internal air removed → collapse), using the notation $P\_out > P\_in \rightarrow$ collapse. Students who finish early are	1. Say: 'Open your cumulative model from Lesson 1. Today's job: add the atmosphere. Draw arrows — lots of them — showing where the atmosphere pushes on the jerrycan. Use the notation we just practised.' Project a model template on the board showing a blank jerrycan outline with a reminder key: OUTSIDE arrows = atmospheric pressure; INSIDE arrows = internal air pressure. 2. Circulate for 7 minutes. Check specifically: (a) Are arrows present on ALL sides including top, not just the sides? (b) Are arrows LABELLED with P_atm or P_out? (c) Does the student's 'collapsed'	STRATEGY — Cumulative Model Revision (Incremental Model-Based Learning): Returning to the same model diagram across every lesson in the sequence embeds NGSS Practice 2 (Developing and Using Models) as a sustained habit rather than a one-off activity. The notation $P\_out > P\_in \rightarrow$ collapse introduces scientific symbolic reasoning without requiring full mathematical formalism yet, bridging qualitative and quantitative understanding. The peer check ('look at your neighbour's model') embeds collaborative accountability and exposes individual misconceptions	INDICATOR: Completed models must show: (1) arrows on at least 4 surfaces (sides, top — bottom optional but accepted), (2) correct labelling of P_atm and P_internal, (3) a clear visual distinction between 'balanced' and 'collapsed' states. Models with arrows only pointing downward indicate a gravity-pressure conflation misconception. Models with no labelling indicate that the student has the visual intuition but lacks the vocabulary to formalise it. Both types of incomplete models are recorded by the teacher for targeted support at the start of Lesson 5. The teacher's 'LESSON

for similar readings	challenged to add a second mini-diagram showing the mercury barometer and labelling where atmospheric pressure acts. The teacher circulates and stamps or initials completed models with a 'LESSON 4 EVIDENCE ADDED' mark to track cumulative revision.	diagram show the inward collapse as a result of P_out > P_in? 3. If a student has only drawn downward arrows (a common misconception — thinking gravity is doing the collapsing), kneel to their level and ask: 'When you swam in the pool/river — did the water only push you down, or did it push from the sides too? Now apply that to air.' Wait 10 seconds for the student to self-correct before intervening further. 4. With 2 minutes remaining, ask the class: 'Hold up your model. Look at your neighbour's. Do you both have arrows on the TOP of the jerrycan?' Pause. 'If your neighbour doesn't — help them add it now.' 5. Close with: 'Your model is becoming more powerful with every lesson. By Lesson 9, your model will answer the ENTIRE driving question — machines, lives, mountains. Keep building.'	(e.g., pressure only from above) in a low-stakes, supportive format.	4 EVIDENCE ADDED' stamp creates a cumulative formative record of each student's model growth across the 9-lesson sequence.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record brief notes in your lesson journal:

1. PHENOMENON CONNECTION: Could every student articulate, by the end of Phase 3, that the jerrycan collapsed because of a pressure IMBALANCE — not because the plastic was weak? If not, which phase broke down — observation, reasoning, or direct explanation?

2. MISCONCEPTION TRACKING: How many students initially had arrows pointing only downward in their cumulative models (conflating gravity with pressure)? What questioning move was most effective in shifting this misconception?

3. EVIDENCE QUALITY: Which of the three evidence sources (collapsing tin, Magdeburg video, barometer diagram) generated the most productive student discussion? Which generated the least? What would you substitute or sequence differently in a future iteration?

4. DQB VITALITY: Are student questions on the revised DQB cards becoming more sophisticated — moving from 'what is pressure?' (definitional) to 'why does pressure change with altitude?' (mechanistic/relational)? If not, what scaffolding is needed to push question quality upward?

5. KENYAN CONTEXT: Did the Lake Victoria swimming analogy effectively bridge the abstract concept of omnidirectional fluid pressure for students who have lived experience near water bodies? For inland students (e.g., from Kericho or Eldoret), what alternative embodied analogy might work better?

6. PACING: The lesson is designed for 80 minutes. If your school runs 40-minute periods, split as follows — Period A: Phases 1 and 2 (Predict + Observe); Period B: Phases 3, 4, and 5 (Explain + DQB + Model). Carry the Prediction Wall sticky notes between periods.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed a metal tin implode dramatically when sealed after steam was generated inside and then cooled rapidly in cold water — a controlled replay of the jerrycan phenomenon. They also watched the Magdeburg Hemispheres video showing teams of horses unable to separate two hemispheres held together solely by atmospheric pressure, and annotated a mercury barometer diagram showing how ~760 mm of mercury is supported by atmospheric pressure alone.
What did I learn?	Students learned that the atmosphere is a fluid with weight, and that this weight creates a pressure of approximately 101.3 kPa acting in ALL directions on every surface it contacts. When the air inside a container is removed, the internal pressure drops, the external atmospheric pressure is no longer balanced, and the container is crushed inward. This is exactly what happened to the jerrycan: it was not weak — the atmosphere is simply enormously powerful, and balance is what kept the jerrycan intact.
How does this explain the phenomenon?	Students explained the jerrycan collapse using the pressure imbalance model: $P_{outside}$ (atmospheric) = P_{inside} (air) in the intact jerrycan, creating equilibrium; when internal air was removed, $P_{outside} > P_{inside}$, and the net inward force from the atmosphere overwhelmed the structural strength of the plastic, causing implosion. Students represented this explanation in their ERC organisers, pressure-balance diagrams, and cumulative jerrycan models with labelled arrows.

LESSON 5 (80 minutes): Transmission of Pressure in Fluids — Pascal's Principle	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and explain how pressure applied to an enclosed fluid is transmitted equally in all directions, and to apply this understanding to hydraulic machines used in Kenya.
Knowledge	By the end of this lesson, the learner should be able to: (a) state Pascal's Principle of pressure transmission in fluids; (b) explain how pressure is transmitted equally in all directions in an enclosed fluid; (c) apply the relationship $F_1/A_1 = F_2/A_2$ to solve problems involving hydraulic systems.
Skills	By the end of this lesson, the learner should be able to: (a) design and carry out a simple investigation to demonstrate pressure transmission in fluids; (b) analyse data from a hydraulic model to calculate force multiplication; (c) construct a labelled diagram of a hydraulic machine used in Kenya.
Attitudes	By the end of this lesson, the learner should be able to: (a) appreciate that understanding fluid pressure transmission has enabled the design of machines that improve lives across Kenya; (b) show curiosity and persistence when investigating how small forces can produce large effects through fluids.
Key Inquiry Question	How does pressure in fluids affect the design of machines and structures?
Purpose in Storyline	In Lessons 1–4, learners established that atmospheric pressure crushed the jerrycan by pushing inward when internal pressure dropped, and that liquid pressure increases with depth and density. This lesson advances the storyline by asking: can pressure be deliberately transmitted through a fluid to do useful work? Learners now discover that the same fluid pressure principle that collapsed the jerrycan can be harnessed — in hydraulic car lifts in Nairobi garages, sugarcane presses in Mumias, and life-saving brake systems on matatus across Kenya. This lesson is the pivotal 'engineering turn' in the unit storyline: pressure stops being something that merely crushes and starts being something humans control.
Safety Notes	Ensure syringes used in investigations have no sharp needle tips — use blunt-tipped syringes only. Water or coloured liquid should be used in hydraulic models; avoid oils that may irritate eyes. Students must not point syringes at faces. Wipe up any spills immediately to prevent slipping. If a commercial hydraulic demonstration kit is used, follow the manufacturer's pressure limits.

B. LESSON OVERVIEW
Learners begin by predicting whether a force applied to one side of a fluid-filled closed system will be felt on the other side — and whether it will be stronger, weaker, or equal. They then gather evidence through a hands-on syringe investigation (two syringes of different sizes connected by a tube filled with coloured water) and a short video clip of a hydraulic car-lift in a Nairobi garage. Using their evidence, they derive Pascal's Principle and apply the equation $F_1/A_1 = F_2/A_2$ to solve real Kenyan-context problems. The lesson closes with learners updating their class model to show how the same atmospheric pressure that crushed the jerrycan can be intentionally transmitted through enclosed fluids to multiply force in hydraulic machines.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two syringes of different sizes (5 mL and 20 mL) connected by a short rubber tube. The tube and both syringes are filled with coloured water (no air bubbles). The system is sealed. Learners are asked to write individual predictions (Think-Write-Pair-Share): (1) 'If I push the small syringe plunger in with one finger, what will happen to the large syringe plunger?' (2) 'If the small syringe has a cross-sectional area of 1 cm^2 and the large one has 4 cm^2, and I push with a force of 2 N on the small one — predict the force that will push out the large plunger.' (3) 'How does this connect to what crushed our jerrycan?' Learners record their numeric predictions and reasoning in their ARES notebooks before any demonstration occurs.	Display the two syringes at the front and ask: 'Before I touch anything — look carefully at these two syringes connected by water. Write down in your notebook: what do you think will happen to the big plunger when I push the small one? Be specific — will it move? Will it push harder, weaker, or the same?' Allow 3 minutes of silent individual writing. Then say: 'Now turn to your partner and compare predictions. Do you agree? Who predicted a BIGGER force coming out of the large syringe? Who predicted smaller? Who predicted equal?' Cold-call 4 pairs (select one agreement pair and one disagreement pair). Record all unique predictions on the board under the heading 'OUR PREDICTIONS — LESSON 5'. Do NOT correct any prediction. Say: 'We are about to find out who is right — and the answer may surprise all of us.'	Think-Write-Pair-Share with Numeric Prediction: Requiring learners to commit to a specific number (e.g., '2 N in → ? N out') before the investigation forces them to access prior knowledge about pressure and area, creates a concrete cognitive anchor, and generates productive disagreement that motivates the observation phase. Recording all predictions publicly on the board builds accountability and makes confirmation/disconfirmation of each prediction visible to the whole class.	Observable indicator: Every learner has a written numeric prediction in their notebook with a stated reason before the demonstration begins. Teacher circulates during silent writing (2 min) and scans for: (a) learners who predict 'equal force' — they likely have not yet connected area to pressure; (b) learners who predict 'larger force from larger syringe' — they are beginning to reason about area-force relationships. Teacher notes 2–3 specific misconceptions to target during the Explain phase.
Observe Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — SYRINGE HYDRAULIC INVESTIGATION (25 min): Learner groups of 4 receive: two syringes (5 mL and 20 mL), a 30 cm rubber/PVC tube, coloured water, a 50 g mass, and a ruler. Groups fill the system with coloured water, removing all air bubbles (critical step — teacher explains why: 'Air is compressible; water is not — this matters greatly for our jerrycan story'). Learners then: (i) Push the small syringe with one finger and observe the large syringe plunger; (ii) Place the 50 g mass on the small plunger and measure how high the large plunger rises; (iii) Reverse — place the mass on the large plunger and observe the small syringe; (iv) Record all	Before Activity A, explicitly connect to the phenomenon: 'Remember our jerrycan — the air OUTSIDE pushed it in because the pressure inside dropped. Today we flip the question: can we deliberately PUSH pressure INTO a fluid and have it appear somewhere else? Let's find out.' Circulate during the syringe activity. At the 8-minute mark, pause all groups and ask: 'Hands up — who has noticed something SURPRISING about the movement of the large plunger?' Cold-call 3 students. If no group has removed air bubbles properly, demonstrate: 'Watch what happens when there is air in the tube' — compress an air-filled tube to show it squishes. 'Now watch the water-filled one.' During	Structured Observation with Comparative Investigation: By having learners reverse the syringes (large input → small output vs. small input → large output), they directly observe that the relationship between force and area is reciprocal — a small area input creates a pressure that produces a large force over a large area. This bidirectional testing prevents the misconception that 'the bigger syringe always wins.' The video grounds abstract investigation results in a real Kenyan industrial context (Nairobi jua kali garage), making the principle immediately relevant and memorable.	Observable indicator: Each group's data table has at least 3 rows of recorded observations with numeric values (force, estimated area, movement). Teacher checks for: (a) groups that have air bubbles still in the tube — redirect immediately as this invalidates results; (b) groups that notice the large plunger moves a SHORTER distance than the small plunger travels — this is a key observation for the Explain phase (conservation of energy); (c) groups that correctly identify that the force OUT is greater than the force IN when pushing the small syringe. Stamp or initial notebooks of groups with complete, accurate data tables.

	observations in a structured table: Input Syringe | Area (cm²) | Force In (N) | Output Syringe | Area (cm²) | Observed Force/Movement Out. ACTIVITY B — VIDEO EVIDENCE (10 min): Learners watch a 4-minute video of a hydraulic car-lift at a jua kali garage in Nairobi lifting a Toyota Noah. Guided viewing questions are on the board: (1) 'What is the mechanic doing to create the input force?' (2) 'How does a small pump create enough force to lift a 1,500 kg vehicle?' (3) 'Where have you seen this principle used elsewhere in Kenya?' Learners write observations in their notebooks.	video, pause at 2:00 and ask: 'What force is the pump applying — is it large or small? Look at the cylinder size.' WAIT 12 seconds. Cold-call 2 students. After video: 'In your notebook, write one sentence connecting this Nairobi garage lift to what we just did with our syringes.'		
Explain Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART A — DERIVING PASCAL'S PRINCIPLE (15 min): Using their syringe data, learners are guided to calculate pressure at the input syringe (P = F/A) and compare it to pressure at the output syringe. They discover that pressure is equal on both sides. Teacher formally states Pascal's Principle. Learners write it in their own words, then compare with the formal statement. They then derive the equation $F_1/A_1 = F_2/A_2$ from the principle. PART B — KENYAN CONTEXT PROBLEM SOLVING (10 min): Learners solve two problems in their notebooks: PROBLEM 1 (Mumias Sugar): 'A hydraulic sugarcane press in Mumias has an input piston of area 0.002 m² and an output piston of area 0.1 m². A worker pushes the input piston with a force of 50 N. What force does the output piston exert on the sugarcane?' PROBLEM 2 (Matatu Brakes): 'The brake master cylinder of a matatu has a piston area of 5 cm². The brake pad	Begin Part A by writing on the board: 'P = F/A — you calculated this in Lesson 3 for liquids. Now calculate P for YOUR syringe data.' Give 4 minutes. Then: 'What do you notice about the pressure values on input and output sides?' WAIT 15 seconds. Cold-call 3 students. Guide toward: 'Pressure is the same throughout the enclosed fluid.' Write on board: PASCAL'S PRINCIPLE — 'Pressure applied to an enclosed fluid is transmitted equally and undiminished in all directions throughout the fluid.' Ask: 'What does UNDIMINISHED mean? Tell your partner.' Cold-call 2 students. For Problem 1, say: 'A worker in Mumias sugar factory — just like pressing with your finger on the small syringe — pushes with 50 N. How much force comes out the other side? Use your principle.' WAIT 10 seconds before accepting hands. After learners attempt: 'Who got an answer bigger than 50 N? Raise your hand. Who got smaller? Explain	Data-to-Principle Induction followed by Application Transfer: Rather than stating Pascal's Principle and then verifying it, learners first calculate pressure from their own experimental data and discover the equality themselves — then the teacher formalises what they found. This inductive approach produces deeper conceptual ownership. The Kenyan context problems (Mumias sugar, matatu brakes) immediately require learners to transfer the principle across domains, testing for flexible understanding rather than rote recall. The final jerrycan reconnection enforces the coherent storyline: the same physics that destroys can be engineered to build.	Observable indicator: Learners can correctly solve at least one of the two context problems (P1 answer: 2,500 N; P2 answer: 720 N) and show their working using $F_1/A_1 = F_2/A_2$. Teacher targets: (a) learners who divide instead of multiply (conceptual error — they think bigger area means less force, not more); (b) learners who cannot connect their syringe data to the equation (ask: 'What was your input area? What was your output area? What did you OBSERVE about the forces?'). Exit-ticket check: teacher reads 3 randomly selected notebooks to verify correct equation setup before moving to DQB.

	cylinder has a piston area of 30 cm^2. If the driver pushes the pedal with a force of 120 N, what force is applied to the brake pad?' Learners check answers with partners before class discussion. PART C — CONNECTING TO THE JERRYCAN (5 min): Class discussion — 'The jerrycan collapsed because pressure from outside exceeded pressure inside. Pascal's Principle says pressure transmits equally through a fluid. How are these two ideas connected? Could we use a hydraulic system to RE-INFLATE the jerrycan?'	your reasoning.' For Part C: 'The jerrycan was destroyed by uncontrolled pressure difference. Engineers in Nairobi use the SAME principle but CONTROL it. Write in your notebook: what is the difference between what happened to the jerrycan and what happens in a hydraulic lift?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Pascal's Triangle - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky notes: (1) GREEN NOTE — 'One thing I can now explain about our driving question that I could NOT explain before Lesson 5.' (2) ORANGE NOTE — 'One new question this lesson has raised for me about pressure in fluids, hydraulic machines, or the jerrycan.' Learners post notes on the class DQB under the columns: 'What We Now Understand' and 'New Questions We Have.' The class then reads all new orange-note questions aloud. As a group, learners vote (show of hands) on the TWO most important new questions to investigate in future lessons. Teacher writes the winning questions on the DQB in a special colour and says: 'These are YOUR questions. We will answer them in Lessons 6 and 7.'	Give exactly 3 minutes for silent sticky-note writing. Circulate — if a learner's green note says something vague like 'I learned Pascal's Law,' prompt: 'Can you be MORE specific? What exactly about the jerrycan or hydraulic machines can you explain now that you couldn't this morning?' After posting, read 4 green notes aloud and ask the class: 'Do you agree this is something we can now explain? Thumbs up or down.' For orange notes, after the vote, explicitly connect to the storyline: 'Notice that our questions are getting MORE specific. In Lesson 1 we asked: WHY did the jerrycan crush? Now we are asking things like: HOW do engineers design brake systems to not fail? This is exactly how scientists think — each answer generates better questions.' Update the master DQB tracking chart on the wall, crossing off any questions from Lessons 1–4 that this lesson has now answered.	Metacognitive DQB Protocol with Collaborative Prioritisation: The two-colour sticky-note system requires learners to simultaneously consolidate learning (green) and generate new questions (orange), developing both knowledge organisation and scientific curiosity as explicit skills. The class vote on future questions gives learners genuine agency over the inquiry direction, increasing motivation. Publicly crossing off answered questions from earlier lessons makes conceptual progress visible and builds learners' confidence in their growing understanding.	Observable indicator: Green notes contain lesson-specific, phenomenon-connected statements (e.g., 'I can now explain why a small pump can lift a matatu — because pressure is transmitted equally through the fluid and the large piston area multiplies the force'). Red-flag indicator: a green note that simply restates the principle without connecting it to either the jerrycan or a Kenyan application — teacher pulls that learner aside for a 1-minute verbal probe: 'Tell me how today's lesson connects to what we saw on Day 1 with the jerrycan.'
Model	Learners return to their cumulative	Before model building begins,	Cumulative Model Revision with	Observable indicator: Each group's

Building Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	unit model (begun in Lesson 1). Each group has a large sheet of paper with their developing model of 'How pressure works.' Today's addition task: ADD a new section to the model labelled 'PRESSURE TRANSMISSION IN ENCLOSED FLUIDS.' The section must include: (1) A diagram of two pistons of different sizes connected by fluid, with labelled forces (F_1, F_2) and areas (A_1, A_2); (2) An arrow showing the direction of pressure transmission and the label 'EQUAL PRESSURE — Pascal's Principle'; (3) A real Kenyan example drawn and labelled (group's choice: hydraulic car lift, matatu brakes, hydraulic sugarcane press, hospital patient-lifting equipment, Nairobi construction crane); (4) A dotted arrow connecting this new section BACK to the jerrycan diagram from Lesson 1, with a written sentence explaining the connection. Groups present their updated model section in 60 seconds each (gallery walk format — groups rotate to read other models and place a star sticker on the clearest diagram they see).	display the Lesson 1 jerrycan diagram at the front: 'Your model started HERE — with a mystery. Look at how much you have added since Lesson 1. Today you add the piece that turns pressure from a destroyer into a tool.' Circulate during model building. Stop at each group and ask one probing question: (Group 1) 'Why must your fluid be liquid, not gas, for Pascal's Principle to work reliably?' (Group 2) 'In your matatu brake diagram — which piston is bigger, the master cylinder or the brake pad cylinder? Why?' (Group 3) 'If there were an air bubble in your hydraulic system, what would happen? Connect this to what you saw in the syringe activity.' During gallery walk, say: 'You have 90 seconds at each model. Write one specific praise and one specific question on the sticky note you leave behind.' Close by asking: 'When we look at the whole model from Lesson 1 to today — what is the BIG IDEA connecting every lesson so far?' WAIT 15 seconds. Cold-call 3 students. Guide toward: 'Pressure in fluids acts in all directions, depends on depth, density, and gravity — and can be transmitted and multiplied. The jerrycan showed us the destructive side. Today we saw the constructive side.'	Cross-Lesson Connection Mapping: Requiring the dotted arrow connecting today's new content BACK to the Lesson 1 jerrycan diagram forces learners to construct an explicit causal chain across the entire unit storyline. This prevents knowledge from being stored as isolated lesson-by-lesson facts and instead builds an integrated, coherent mental model. The gallery walk with peer annotation (star sticker + written note) introduces peer assessment as a sensemaking tool and exposes learners to multiple valid representations of the same concept.	model update contains all four required elements. Teacher specifically checks the dotted arrow + written sentence connecting to the jerrycan — this is the highest-order requirement. Sentence quality rubric (applied mentally during circulation): Level 1 — 'They are both about pressure' (insufficient); Level 2 — 'The jerrycan was crushed by transmitted atmospheric pressure; hydraulic machines use transmitted fluid pressure to multiply force' (acceptable); Level 3 — 'Both the jerrycan and hydraulic machines demonstrate Pascal's Principle — pressure is transmitted equally through enclosed fluids — but in the jerrycan the pressure difference was uncontrolled and destructive, while in hydraulic machines engineers deliberately create and control the pressure difference to do useful work' (excellent). Teacher notes which groups achieve Level 3 for recognition at the start of Lesson 6.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions in your ARES teaching journal: (1) CONCEPTUAL CLARITY — Did learners successfully distinguish between 'pressure is equal throughout the fluid' (Pascal's Principle) and 'force is multiplied because of area difference'? Many learners conflate these. If not, plan a 5-minute re-teach opener for Lesson 6 using the syringe model again. (2) PHENOMENON CONNECTION — When learners wrote their DQB green notes, did they explicitly reference the jerrycan, or did the lesson feel disconnected from the anchoring phenomenon? If disconnected, strengthen the 'Part C' jerrycan reconnection in the Explain phase next time by making it a 10-minute structured discussion rather than 5 minutes. (3) EQUITY CHECK — Were girls equally cold-called as boys during the Predict and Explain phases? Review your cold-call tally. Physics classrooms in Kenya often show gender participation gaps — deliberately call on female learners for the highest-cognitive-demand questions (e.g., the conservation of energy implication when the large piston moves less distance). (4) PROBLEM SOLVING — How many learners correctly solved both Kenyan context problems independently? If fewer than 60% solved Problem 2 (matatu brakes) correctly, consider adding a unit-conversion scaffolding step (cm^2 to m^2) at the start of Lesson 6. (5) MODEL QUALITY — Review 3–4 group models. Are the connections between lessons becoming richer and more specific, or are models growing in size without growing in depth? If models are superficial, introduce a 'minimum sentence standard' for model annotations in Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners pushed a small syringe connected by a water-filled tube to a large syringe and observed the large plunger produce a greater force than the input force. They also watched a hydraulic car-lift at a jua kali garage in Nairobi raise a full Toyota Noah vehicle using a small pump. They recorded that removing air bubbles from the syringe system was essential for reliable pressure transmission.
What did I learn?	Learners learned that pressure applied to an enclosed fluid is transmitted equally and undiminished in all directions (Pascal's Principle). They derived and applied the equation $F_1/A_1 = F_2/A_2$ to show that a small force on a small piston creates the same pressure as a large force on a large piston. They connected this to real Kenyan hydraulic machines: sugarcane presses in Mumias, matatu brake systems, and hydraulic lifts in Nairobi garages.
How does this explain the phenomenon?	Learners explained that the same pressure-transmission principle that allowed the atmosphere to crush the jerrycan (uncontrolled pressure difference) is deliberately harnessed in hydraulic machines (controlled pressure difference). The jerrycan collapsed because external atmospheric pressure exceeded internal pressure; hydraulic machines exploit the fact that pressure is equal throughout an enclosed fluid — so by choosing output piston areas much larger than input piston areas, engineers multiply force to lift heavy loads, press sugarcane, and stop matatus safely.

LESSON 6 (80 minutes): Liquid Pressure: Depth, Density, and the Formula P = ρgh	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to investigate and explain how liquid pressure depends on depth and density, and to apply the equation $P = \rho gh$ to solve problems related to pressure in fluids.
Knowledge	Learners will know that liquid pressure increases with depth (h) and with the density (ρ) of the liquid, and that it acts equally in all directions at a given depth. Learners will know the equation $P = \rho gh$ and the meaning of each variable.
Skills	Learners will design and conduct a fair experiment using modified plastic bottles filled with water, record data in structured tables, identify patterns from data, and apply $P = \rho gh$ to calculate liquid pressure at different depths.
Attitudes	Learners will appreciate the importance of systematic investigation in generating reliable evidence, and will value collaborative scientific inquiry as a means of understanding natural phenomena.
Key Inquiry Question	How does pressure in fluids affect the design of machines and structures?
Purpose in Storyline	In Lesson 5 students established that atmospheric pressure is real and enormous — sufficient to crush the jerrycan. Now, learners shift their gaze inward: if the jerrycan had remained full of water from Lake Naivasha, what pressure would the water itself exert on the walls? This lesson investigates liquid pressure as a function of depth and density, laying the quantitative foundation ($P = \rho gh$) that will later explain hydraulic machines, dam design, and submarine structure.
Safety Notes	Ensure plastic bottles have no sharp edges after modification. Wipe up any water spills immediately to prevent slipping. Students should wear closed shoes during the practical. Do not use glass containers. Supervise the use of rulers and sharp tools when making holes in bottles.

B. LESSON OVERVIEW
Learners begin by predicting how the speed of water squirting from holes at different heights on a tall plastic bottle will vary — connecting to the crushed jerrycan context. They then conduct a structured investigation using modified 2-litre bottles (representing sections of a tall water column, as if sourced from Lake Naivasha's water system) with holes at measured depths, observing jet distance as a proxy for pressure. A class data table is built collectively, patterns are identified, and the equation $P = \rho gh$ is introduced through guided sensemaking. Learners then apply the formula to calculate pressures at the base of real Kenyan water towers and dam walls. The lesson closes with a revised Driving Question Board entry and model update linking liquid pressure to the jerrycan scenario.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12	Each learner receives a sketch of a tall transparent tube sealed at the bottom with three holes drilled at different heights: near the top (shallow), at the middle, and near	Distribute the sketch diagram. Say: 'Look carefully at this tube filled with water — three holes, three depths. Before you see anything, I want your best scientific prediction.	Prediction Sorting Wall — learners' sticky-note predictions are publicly displayed and sorted before the experiment. This activates prior intuitions about gravity and water,	Teacher scans written predictions for the quality of reasoning (not correctness). Observable indicator: learner can articulate at least one causal factor (e.g., 'because there

🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	the bottom (deep). The tube is filled with water. Learners individually write a prediction: 'Which hole will shoot water the farthest? Rank them 1 (farthest) to 3 (shortest). Write one sentence explaining your reasoning.' Learners then share predictions with a shoulder partner before the teacher cold-calls five students. Predictions are posted on sticky notes on the board under the headers: SHALLOW / MIDDLE / DEEP — sorted by which hole students predict shoots farthest.	Write it down individually — silence for two minutes.' [WAIT 2 minutes.] 'Now share with your shoulder partner — you have 90 seconds.' [WAIT 90 seconds.] Cold-call exactly FIVE students using name cards: 'Akinyi — what did you predict, and why?' 'Mutua — do you agree or disagree? What's your reasoning?' Tally predictions on the board. Say: 'Notice we have disagreement in this room — that is exactly what science looks like before the evidence arrives. Let's find out who the data agrees with.'	creates a testable record to return to, and generates productive cognitive dissonance when results contradict majority predictions. It primes learners to notice the relationship between depth and pressure.	is more water above it pushing down') even if the prediction is incorrect. Teacher notes which students invoke gravity/weight versus those who predict all holes shoot equally — these are the key misconceptions to address in Phase 3.
Observe Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In groups of four, learners conduct the 'Lake Naivasha Tower Investigation.' Each group has: one 2-litre plastic bottle with three pre-drilled holes sealed with tape at depths 5 cm, 15 cm, and 25 cm from the base of a water column; a ruler; a large tray or basin; a data table printed on A3 paper. Procedure: (1) Fill the bottle with water up to the brim. (2) Hold a ruler flat on the tray surface. (3) Unseal each hole one at a time (re-tape between trials) and measure the horizontal distance (cm) the water jet travels before hitting the tray. (4) Repeat three times per hole and record all values. (5) Groups then switch one variable: they repeat using a saltwater solution (to approximate denser liquid — dissolve 4 tablespoons of salt in the water) and record jet distances. A full class data table is constructed on the board as group recorders call out their values. Teacher leads the class in calculating mean jet distances per hole depth for both fresh and salt water.	Before groups begin, say: 'Before you unseal any tape — tell me: what is your ONE variable changing in the first set of trials? [WAIT 10 seconds. Cold-call 2 students.] What are you keeping the same? [WAIT 10 seconds. Cold-call 2 different students.]' Circulate during the investigation. Pose probing questions: 'Kamau, your group got different distances for the same hole — why might that happen?' 'Njeri, you said the bottom hole shot the farthest — is that what you predicted? What does your group think explains it?' After all groups record data, stand at the board and say: 'Recorders, call out your mean jet distance for hole at 5 cm depth.' Record all groups' values in the class table. Say: 'Look at the class data. What pattern do you see? Write one sentence in your notebook — silent writing, 60 seconds.' [WAIT 60 seconds.] Cold-call 3 students.	Fair-Test Investigation with Class Data Aggregation — individual group results are pooled into a whole-class data set, modelling real scientific practice where multiple trials reduce error. The use of salt water as a second condition introduces density as a second variable in a concrete, observable way, directly setting up the two factors (h and ρ) in $P = \rho g h$ before any formula is introduced.	Observable indicators: (1) Each group's data table shows three trials per hole with a calculated mean — if groups only have one reading per hole, teacher intervenes and asks 'Why do scientists repeat measurements?' (2) During class data sharing, teacher checks whether groups can read their own tables accurately. (3) The 60-second sentence written after data compilation should mention 'deeper' or 'bottom' and 'farther' — any student who writes the opposite should be flagged for targeted questioning in Phase 3.

Explain Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher uses a worked 'Naivasha Water Tower' context to introduce the equation P = ρgh. Learners first attempt a 'column of water argument': they are shown a diagram of a 1 cm^2 cross-section column of water at three depths. Using the idea that more water sits above a deeper point, they reason that weight (and thus force) increases with depth. The teacher then writes P = ρgh on the board and introduces each variable with a Kenyan anchor: ρ (density of water = 1000 kg/m^3 — 'the same water carried in jerrycans in Kibera'), g (gravitational field strength = 10 N/kg — 'the same g that pulls maize down on Rift Valley farms'), h (depth in metres — 'measured from the water surface, like the depth of Lake Victoria at 84 m'). Learners work in pairs on three graduated problems: (a) Calculate pressure at the base of a 3 m tall water tank in Kericho. (b) Calculate pressure at h = 84 m in Lake Victoria ($ρ_{fresh}$ = 1000 kg/m^3). (c) Calculate pressure at the base of a saltwater column of 2 m depth ($ρ_{salt}$ = 1025 kg/m^3) — compare with the freshwater result and explain why they differ. Groups then return to their jet-distance data: 'Does your data agree with what P = ρgh predicts? Use the formula to calculate the pressure at each of your three hole depths and rank them. Does the ranking match the jet distance ranking?'	Draw the column diagram on the board. Say: 'Imagine a tiny square — 1 cm by 1 cm — at the bottom of a water column. At 5 cm depth, how many centimetres of water sit above it? [WAIT 10 seconds.] At 25 cm depth? [WAIT 10 seconds.] Which tiny square has more water pushing down on it?' Cold-call 3 students. Write P = ρgh. Say: 'This formula captures exactly what you just reasoned. Let me unpack each letter.' Write each variable with its unit and Kenyan anchor as described. 'Now, pairs — attempt problem (a). Show every step: write the formula, substitute values, calculate, write units. Five minutes.' [WAIT 5 minutes.] Circulate — look for unit errors. Cold-call one pair to present at the board. For problem (c) say: 'Compare your freshwater and saltwater answers — what is different, and why? Write one sentence. [WAIT 60 seconds.] This should remind you of what happened when your group used salt water in the bottle experiment — did the jet go farther or shorter? Does P = ρgh predict that?' Cold-call 2 students.	Anchored Formula Introduction + Evidence Reconciliation — the formula is introduced after learners have built an intuitive argument from the column diagram, not before. The 'Does your data agree?' task requires learners to use the formula to generate predictions and compare them against their own experimental results, closing the loop between empirical evidence and theoretical explanation. This is NGSS SEP 5 (Using Mathematics and Computational Thinking) and SEP 6 (Constructing Explanations).	Observable indicators: (1) In problem (a), learners should write P = ρgh = 1000 × 10 × 3 = 30,000 Pa — teacher checks for correct unit (Pa). (2) In problem (c), learners should identify that denser liquid (salt water) produces greater pressure at the same depth. (3) In the evidence reconciliation task, learners should confirm that their hole-depth ranking (deep = highest pressure = longest jet) matches the formula's prediction. Any group whose ranking does not match should re-examine their experimental procedure. Exit check: teacher asks three students at random — 'Name the three factors that affect liquid pressure' — all three must say depth, density, and gravitational field strength.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Pascal's Triangle	Each learner writes on a sticky note: one thing they can now answer on the Driving Question Board, and one new question today's lesson raised. The class DQB is updated: the teacher	Hand out sticky notes. Say: 'Two minutes — one thing you can now explain, one new question today raised. Silent writing.' [WAIT 2 minutes.] Invite volunteers to the DQB. Read aloud the question	Driving Question Board Column Migration — physically moving questions from 'unknown' to 'evidenced' makes the growth of understanding visible and cumulative. The bridging prompt	Observable indicators: (1) The new sticky-note question demonstrates conceptual progression — not a repeat of earlier questions, but one that extends today's learning (e.g.,

- Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	invites two volunteers to physically move previously posted questions from the 'We don't know yet' column to the 'We now have evidence for this' column, reading the original question aloud and explaining in one sentence what today's lesson revealed. New questions are added. The teacher then poses a bridging prompt: 'We now know the pressure at the base of a water-filled jerrycan. But the jerrycan we watched was empty and sealed — and outside air was pressing in. How does today's formula help us understand why the outside air could crush the walls?' Learners write a two-sentence bridge in their notebooks.	being moved: 'Does the depth of the water matter for how hard it pushes? — what's our evidence today?' Prompt the volunteer: 'Read the question, say what we found, and move the card.' After DQB update, pose the bridging prompt. Say: 'Think about the crushed jerrycan. The water we measured today pushes outward on the bottle walls. Atmospheric pressure pushes inward on the jerrycan walls. Both can be calculated with $P = \rho gh$ — or for air, with our earlier atmospheric pressure knowledge. Write two sentences connecting today's formula to the jerrycan phenomenon.' [WAIT 2 minutes.] Cold-call 2 students to share.	requires learners to self-connect new quantitative knowledge ($P = \rho gh$) back to the anchoring phenomenon, preventing siloing of lesson content and reinforcing storyline coherence.	'What happens to pressure when a diver goes deeper in Lake Tanganyika?'). (2) The two-sentence bridge should mention both liquid pressure (outward) and atmospheric pressure (inward) and the idea of pressure difference. Teacher collects notebooks from 4 randomly selected learners to check the bridge sentences for quality.
Model Building Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative pressure model from Lesson 1 (initially a simple annotated drawing of the jerrycan) and add a new layer. Using the provided model template, they draw: (1) a cross-section of the jerrycan when it was full of water from Lake Naivasha — annotating arrows showing water pressure acting outward on the walls at three depths, labelling the bottom arrow as 'highest pressure ($P = \rho gh$, h is greatest here)'; (2) a second diagram of the empty, sealed jerrycan — annotating atmospheric pressure arrows pressing inward from all sides, with a note: 'No liquid inside to push back → net inward force → jerrycan crumples'; (3) a written model statement: 'Liquid pressure depends on depth (h), density (ρ), and gravitational field strength (g). The deeper the liquid and the denser it is, the greater the pressure it exerts outward. When a jerrycan is	Distribute the model templates and checklist. Say: 'Take out your cumulative model. Today we are adding the liquid pressure layer — a new piece of the jerrycan story. You have eight minutes to draw and annotate both diagrams and write your model statement.' [WAIT 8 minutes.] Circulate — prompt: 'Show me where on your diagram the pressure is greatest. Label it with the formula.' After drawing: 'Swap with your partner — use the checklist. Give one specific strength — quote something from their diagram — and one suggestion.' [WAIT 3 minutes.] Close by cold-calling one pair: 'Read your partner's model statement aloud and tell us one thing they explained well.' Conclude: 'Next lesson, we will investigate how this pressure is transmitted through fluids — and how that transmission is used in machines that carry tonnes of maize harvest in Nakuru.'	Cumulative Model Revision with Peer Review — the model is not recreated each lesson but revised and extended, making conceptual growth visible over time. The two-diagram structure (full jerrycan vs. empty sealed jerrycan) directly maps the lesson's new knowledge onto the anchoring phenomenon, ensuring that $P = \rho gh$ is not an abstract formula but a tool for explaining a real, observed event. Peer review using a structured checklist develops scientific communication skills (NGSS SEP 8: Obtaining, Evaluating, and Communicating Information).	Observable indicators: (1) Both diagrams are present and correctly labelled — the full-jerrycan diagram shows outward pressure arrows increasing with depth; the empty-jerrycan diagram shows inward atmospheric pressure arrows. (2) The model statement mentions all three variables (h, ρ, g). (3) The peer-review checklist is completed with specific, evidence-referenced feedback (not vague comments like 'good job'). Teacher collects four random models to assess before Lesson 7.

	emptied and sealed, liquid pressure (outward) is removed, but atmospheric pressure (inward) remains — so the walls collapse inward.' Pairs swap models and use a provided peer-review checklist to give one 'strength' and one 'suggestion for improvement.'			

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners correctly identify depth and density as the key factors before the formula was introduced — or did most wait for the equation? If learners could not reason from the column diagram, re-teach the 'weight of water above' conceptual argument before moving to transmission of pressure in Lesson 7. (2) Were there persistent misconceptions that liquid pressure depends on the volume or shape of the container (the hydrostatic paradox)? If so, bring a narrow tube and a wide container filled to the same height and compare pressures at the base — a powerful corrective demonstration. (3) How accurately did learners reconcile their experimental jet-distance data with $P = \rho gh$ predictions? If the reconciliation was weak, consider adding a brief 'data analysis clinic' at the start of Lesson 7. (4) Check the four collected cumulative models: can learners correctly contrast outward liquid pressure and inward atmospheric pressure in the jerrycan scenario? This dual-pressure understanding is the conceptual hinge for Pascal's Principle in Lesson 7. (5) Note which students were consistently silent during cold-calling — identify them for targeted small-group support in the next lesson.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Water squirting from holes at different depths in a plastic bottle travelled different distances — the deepest hole shot water the farthest. Salt water jets travelled farther than fresh water jets from the same depth.
What did I learn?	Liquid pressure increases with depth (h) and with the density (ρ) of the liquid, and can be calculated using $P = \rho gh$, where g is the gravitational field strength (10 N/kg). Pressure acts in all directions at any given depth.
How does this explain the phenomenon?	When the jerrycan was full of water, liquid pressure pushed outward on all walls — greatest at the base. When emptied and sealed, that outward liquid pressure was removed. Atmospheric pressure (inward) remained unopposed, so the walls were crushed inward. $P = \rho gh$ shows us exactly how large that liquid pressure was — and why its absence matters.

LESSON 7 (80 minutes): Pascal's Principle & Hydraulic Systems — Pressure Transmitted, Power Multiplied	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To develop learners' understanding of how pressure applied to an enclosed fluid is transmitted equally in all directions (Pascal's Principle), and to apply this principle to explain the design and operation of hydraulic machines used across Kenya.
Knowledge	Learners will know that: (i) pressure applied to a confined fluid is transmitted undiminished in all directions (Pascal's Principle); (ii) hydraulic systems use the equation $F_1/A_1 = F_2/A_2$ to multiply force; (iii) hydraulic machines (car brakes, excavators, hospital beds, sugar-cane presses in Mumias) rely on Pascal's Principle; (iv) the jerrycan collapse is the inverse of Pascal's Principle — external pressure transmitted inward by the atmosphere was unresisted because internal pressure was removed.
Skills	Learners will be able to: (i) state and explain Pascal's Principle using fluid-pressure reasoning; (ii) apply $F_1/A_1 = F_2/A_2$ to solve quantitative problems on hydraulic systems; (iii) construct a simple syringe-based hydraulic model and observe force multiplication; (iv) draw and label a hydraulic system diagram; (v) connect Pascal's Principle back to the jerrycan phenomenon.
Attitudes	Learners will appreciate: (i) the elegance of harnessing invisible fluid pressure to accomplish large mechanical tasks; (ii) how Kenyan engineers and technicians apply Pascal's Principle daily in agriculture, medicine, and transport; (iii) the collaborative nature of scientific sensemaking as they build on seven lessons of cumulative evidence.
Key Inquiry Question	How does pressure in fluids affect the design of machines and structures?
Purpose in Storyline	Lesson 7 of 9 — TRANSMISSION UNLOCKED: By Lesson 7, learners have established that atmospheric pressure crushed the jerrycan (Lessons 1–2), that $P = \rho gh$ governs liquid pressure (Lessons 3–4), and that pressure acts in all directions (Lessons 5–6). Now they ask: can pressure applied at ONE point in a fluid do useful work at ANOTHER point far away? Pascal's Principle answers this definitively and explains how the same physics that crushed the jerrycan is harnessed in hydraulic brakes, Mumias sugar presses, and Nairobi hospital lifting beds — turning an invisible force into a mechanical marvel.
Safety Notes	– Syringes — use only plastic medical/veterinary syringes (no needles). Dispose of responsibly after the lesson. • Tubing — ensure all connections are secure before pressurising to avoid sudden disconnection and fluid spray. • Coloured water — use food colouring only; avoid dyes that stain skin. Wipe spills immediately to prevent slipping. • No glass apparatus to be used in the syringe model activity. • Learners with latex sensitivities should use nitrile or vinyl gloves if handling rubber tubing.

B. LESSON OVERVIEW

Learners begin by predicting whether a small force applied on a small syringe can lift a heavy load on a large syringe. They then build a paired-syringe hydraulic model using plastic syringes and flexible tubing filled with coloured water, physically experiencing force multiplication. A structured data-collection activity confirms the $F_1/A_1 = F_2/A_2$ relationship. A short video clip of a Nairobi garage mechanic operating a hydraulic car lift anchors the real-world application. Learners then use Pascal's Principle to re-explain the jerrycan phenomenon — the atmosphere is a vast hydraulic system that transmitted pressure undiminished onto every square centimetre of the jerrycan's surface the moment internal pressure was lost. The lesson closes with model revision on the class Driving Question Board and individual hydraulic-system diagram tasks.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Pascal's Triangle - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a 'Prediction Card' showing a diagram of two syringes connected by tubing. Syringe A has a cross-sectional area of 1 cm^2 and a 2 N force is applied to its plunger. Syringe B has a cross-sectional area of 4 cm^2. Learners individually write: (a) Will the plunger of Syringe B move? (b) Will it push with more force, less force, or the same force as Syringe A? (c) Connect your prediction to the jerrycan — if the atmosphere is 'Syringe A', what is the jerrycan's wall? Learners share predictions with a shoulder partner for 2 minutes, then the teacher cold-calls 4 learners (2 who predict MORE force, 2 who predict SAME OR LESS) to voice their reasoning aloud. Predictions are recorded on sticky notes and posted on the Driving Question Board under 'We think...'	Distribute Prediction Cards as learners settle. Say: 'Before we touch a single syringe, I want your brain's best guess — commit to it in writing. No wrong answers yet.' Allow 3 minutes of silent individual writing. Then say: 'Turn and tell your partner your prediction AND the one sentence of reasoning behind it.' [WAIT 2 minutes.] Cold-call exactly 4 learners — choose deliberately: two who predict force multiplication, two who are sceptical. Ask each: 'What is your prediction AND what evidence from our last six lessons supports it?' [WAIT TIME: 12 seconds per learner before prompting.] Record the four predictions verbatim on the board in two columns: 'Force increases' vs 'Force stays same or decreases'. Say: 'By the end of this lesson, one column will be much richer than the other — let's find out which.'	Predict-Then-Commit (PTC): Requiring written, committed predictions before observation activates prior knowledge from Lessons 1–6, surfaces misconceptions about force and pressure, and creates the cognitive tension needed for genuine learning during the Observe phase. Connecting the prediction explicitly to the jerrycan keeps the anchoring phenomenon alive.	Observable indicator: Every learner has a completed Prediction Card with a written justification referencing pressure concepts (not just a guess). Teacher scans cards during partner talk — flag learners who conflate force and pressure (e.g., 'more area means more force because there is more pressure') for targeted questioning during the Explain phase. Cold-call responses reveal whether learners are reasoning from $P = F/A$ or from intuition alone.
Observe Phase 📹 **VIDEO:** [Pascal's Triangle - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	ACTIVITY A — Syringe Hydraulic Model (15 min): Groups of 4 receive: two plastic syringes of different sizes (e.g., 5 ml and 20 ml), ~40 cm of flexible plastic tubing, coloured water (maji ya rangi), a small 200 g mass (or a stone of known mass), and a data table. Groups fill the system with coloured water (no air bubbles), connect syringe to syringe, and mount the large syringe vertically. They place the 200 g mass on the large syringe plunger, then push the small syringe plunger with one finger. They observe: (i) does the large plunger rise? (ii) does it lift the mass? (iii) how far does the small plunger move compared to the large plunger? They record	Before Activity A, say: 'Your job is not just to make it work — your job is to NOTICE what surprises you and write it down.' Circulate every 90 seconds. At the 5-minute mark, stop all groups and ask: 'Raise your hand if the large plunger moved — now keep your hand up if it surprised you HOW MUCH it moved.' [Count hands aloud.] Ask one surprised group: 'Describe what you saw in one sentence using the word PRESSURE.' [WAIT 10 seconds.] For groups struggling with air bubbles, say: 'Air is compressible — water is not. What does that tell you about why we use water in this system?' Before Activity B, say: 'Watch this mechanic in Industrial Area,	Hands-On Analogical Modelling: The syringe system is a physical analogue of all hydraulic machines. By manipulating the model themselves, learners build embodied understanding of force multiplication and pressure transmission. The structured viewing guide bridges the physical model to industrial scale, and the explicit 'connect to the jerrycan' prompt sustains storyline coherence across all seven lessons.	Observable indicators: (i) Data tables show qualitative observation that small plunger depression → large plunger rise, AND quantitative observation that the force recorded on the spring balance is less than the weight lifted (force multiplication). (ii) Viewing guides contain the words 'pressure transmitted' and a coherent jerrycan connection. Teacher listens for groups saying 'the fluid pushed everywhere equally' — this is the target language of Pascal's Principle emerging organically before formal instruction.

	qualitative observations and measure the force required (using a spring balance on the small plunger) vs the weight lifted (mg of the mass). ACTIVITY B — Video Anchor (7 min): Teacher plays a 4-minute clip of a hydraulic car lift in a Nairobi garage (or a Mumias sugar-cane hydraulic press). Learners complete a structured viewing guide: 'What fluid is used? Where is pressure applied? Where does the output force act? What would happen if there were a leak — connect this to the jerrycan.'	Nairobi. I want you to find Pascal's Principle hiding inside every move he makes.' Pause the video at the moment the lift rises and cold-call 2 learners: 'Where is the small piston and where is the large piston in what you just saw?' [WAIT 12 seconds each.]		
Explain Phase 📹 **VIDEO:** Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	PART A — Formalising Pascal's Principle (8 min): Teacher leads a class discussion to name and state Pascal's Principle verbatim: 'Pressure applied to a confined fluid is transmitted undiminished in all directions throughout the fluid.' Learners write this in their notebooks and immediately annotate their syringe diagram with arrows showing equal pressure transmission in all directions. PART B — Deriving the Hydraulic Equation (7 min): Teacher guides derivation: Since $P_1 = P_2$, then $F_1/A_1 = F_2/A_2$, therefore $F_2 = F_1 \times (A_2/A_1)$. Learners work through two Kenya-context problems: (i) A mechanic in Mombasa's Changamwe garage uses a hydraulic lift. The small piston has area 0.005 m². He applies 500 N. The large piston has area 0.1 m². What force lifts the car? (ii) A Mumias sugar-cane press applies 200 N on a 0.002 m² piston. What force acts on the 0.05 m² pressing plate? PART C — Reconnecting to the Jerrycan (7 min): Teacher facilitates a class discussion using the prompt: 'The atmosphere is the ultimate	For Part A, write Pascal's Principle on the board and say: 'Say it with me.' [Class reads aloud once.] Then say: 'Now close your book and say it to your partner.' [WAIT 20 seconds.] Cold-call 1 learner to say it from memory. For Part B, work the Mombasa problem on the board step by step, narrating each algebraic step. Then say: 'The Mumias problem is yours — you have 4 minutes, working individually, silent.' [WAIT full 4 minutes. Circulate and note errors.] Cold-call 2 learners to write their working on the board. Say to the class: 'If your answer matches, put a tick. If it does not, write one question you have.' For Part C, say: 'I want you to imagine you ARE the jerrycan wall — you are one square centimetre of plastic. Describe your experience using Pascal's Principle.' [WAIT 12 seconds.] Cold-call 3 learners. Listen for the phrase 'undiminished in all directions'. If absent, probe: 'Did the pressure act on just one side of the jerrycan or on every side equally — what does Pascal tell us?'	Derivation + Contextualised Problem Solving + Analogical Narrative: The derivation from first principles ($P_1 = P_2$) connects the new equation to the pressure definition learners already hold. Kenya-context problems (Changamwe, Mumias) make the mathematics feel purposeful. The 'You ARE the jerrycan wall' narrative is a perspective-taking strategy that forces learners to apply Pascal's Principle from inside the phenomenon, deepening conceptual ownership.	Observable indicators: (i) Correct numerical answers to both problems — target: F = 10,000 N (Mombasa) and F = 5,000 N (Mumias). (ii) Three-sentence jerrycan explanations must contain: Pascal's Principle by name, 'undiminished', and a reference to equal external vs internal pressure. Teacher collects 6 randomly selected notebooks at the end of Part C and checks for these three elements before Phase 4 begins.

	hydraulic system. When we removed air from the jerrycan, we removed the internal fluid pressure. The external atmospheric pressure — transmitted equally in ALL directions — had no equal and opposite pressure to balance it. Pascal's Principle tells us that pressure is transmitted undiminished — so every square centimetre of the jerrycan wall received the full 101,325 Pa of atmospheric pressure simultaneously. The jerrycan did not stand a chance.' Learners write a 3-sentence explanation in their own words.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Pascal's Triangle - Overview] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Pressure versus Temperature (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives three colour-coded cards: GREEN ('We now know...'), ORANGE ('This changes our earlier idea that...'), RED ('We still wonder...'). Groups spend 4 minutes writing one card of each colour, then post them on the class Driving Question Board in the 'Lesson 7 — Pascal's Principle' column. A gallery walk (3 minutes) lets all learners read other groups' cards. Each learner then individually writes one sentence on a sticky note beginning: 'Pascal's Principle helps explain the jerrycan because...' and posts it on the board. Teacher reads 4–5 sticky notes aloud, pausing after each to let the class affirm, challenge, or extend.	Say: 'Our Driving Question Board is a living record of our growing understanding. Seven lessons ago, none of us could explain the jerrycan. Look at how far we have come.' Point to the board's progression from Lesson 1 to Lesson 7. Distribute colour-coded cards and say: 'GREEN is for certainty, ORANGE is for revision, RED is for honest confusion. All three are equally valuable in science.' During the gallery walk, say: 'Read silently. Do not talk yet.' [WAIT the full 3 minutes.] After reading sticky notes aloud, ask the class: 'Which of these sticky notes do you think is the strongest scientific explanation — and why?' Cold-call 2 learners. [WAIT 12 seconds each.] Say: 'We have two lessons left. What does our RED column tell us we must still explain?'	Driving Question Board — Cumulative Storyline Tracking: The DQB makes the learning progression visible and public. Colour-coding forces metacognitive sorting (certainty vs revision vs open questions), and the gallery walk builds scientific community norms. The sticky-note sentence starter ('Pascal's Principle helps explain the jerrycan because...') requires learners to synthesise new knowledge with the anchoring phenomenon, maintaining storyline coherence.	Observable indicators: (i) GREEN cards contain accurate statements of Pascal's Principle and/or the hydraulic equation. (ii) ORANGE cards show genuine revision of prior ideas (e.g., 'We thought pressure only pushed down — now we know it pushes in all directions'). (iii) RED cards reveal productive remaining questions (e.g., 'How do hydraulic brakes not fail when there is a small leak?'). Teacher photographs the DQB at the end of the lesson for tracking across the unit.
Model Building Phase 📹 **VIDEO:**	Learners individually draw and annotate a hydraulic system diagram in their exercise books. The diagram must include: (i) a	Say: 'Every scientist keeps a model — a picture of how they think the world works. Today's model must show Pascal's	Cumulative Scientific Modelling: Requiring learners to draw a hydraulic system AND the jerrycan in a single integrated diagram	Observable indicators: (i) Diagrams contain all five required elements — teacher checks during circulation using a 5-point mental

Pascal's Triangle - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	small input piston with F_1 and A_1 labelled; (ii) a large output piston with F_2 and A_2 labelled; (iii) arrows of equal length inside the fluid showing pressure transmitted equally in all directions; (iv) the equation $F_1/A_1 = F_2/A_2$ written beside the diagram; (v) a mini inset panel showing the jerrycan — with 'Atmosphere = giant hydraulic system' annotated, and arrows showing external atmospheric pressure pressing inward unresisted. Below the diagram, learners write a 'Model Update Statement': 'My model of why the jerrycan crushed itself has changed/grown because...' Learners who finish early extend by adding a second diagram showing a hydraulic braking system in a matatu and labelling where Pascal's Principle acts.	Principle AND the jerrycan in the same drawing. They are the same physics.' Give learners 10 minutes of quiet individual work. Circulate and look for: arrows that are all the same length inside the fluid (equal pressure in all directions) — if arrows are different lengths or only pointing downward, say quietly: 'Your arrows are telling me pressure is stronger in one direction — does Pascal's Principle agree with that? Fix it.' [WAIT for learner self-correction.] At the 10-minute mark, ask 3 learners to display their diagrams under the document camera (or hold them up). For each, ask the class: 'What is strong about this model? What is missing?' [WAIT 10 seconds after each display.] Close by saying: 'Your model is now holding seven lessons of evidence. In Lesson 8, we add the final pieces — Bernoulli's Principle and moving fluids. Keep your RED questions alive.'	demands that they unify the two contexts under one principle. The 'Model Update Statement' is a metacognitive writing strategy that makes learning growth explicit. Peer critique via document-camera display builds scientific argumentation skills (NGSS SEP 6: Constructing Explanations) and models the iterative nature of scientific model revision.	checklist. (ii) Arrows inside the fluid are equal length and point in multiple directions (not only downward). (iii) Jerrycan inset clearly labels external pressure and absent internal pressure. (iv) Model Update Statements are collected and sorted into three piles: 'Fully meeting expectations', 'Developing', 'Needs support' — this sorting informs differentiation in Lessons 8 and 9.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PREDICTION QUALITY — Did learners' Prediction Cards show reasoning from pressure concepts (F/A) or from everyday intuition? If most predictions were purely intuitive, consider adding a brief $P = F/A$ review warm-up at the start of Lesson 8. (2) SYRINGE ACTIVITY — Did all groups successfully eliminate air bubbles and achieve a working hydraulic model? If not, consider pre-assembling one demonstration model for groups that struggled, and exploring whether the issue was procedural (tubing connections) or conceptual (why air undermines the system). (3) PROBLEM-SOLVING — Were learners able to apply $F_1/A_1 = F_2/A_2$ independently on the Mumias problem? If more than 30% of learners made errors, identify whether the error was algebraic (rearranging the equation) or conceptual (confusing force and pressure). Address this explicitly in the Lesson 8 warm-up. (4) DQB RED CARDS — Read all RED cards carefully tonight. Do they reveal productive scientific curiosity (leaks in hydraulic systems, Bernoulli, moving fluids) or fundamental gaps (still unclear on Pascal's Principle itself)? The RED cards should directly shape the opening of Lesson 8. (5) JERRYCAN CONNECTION — In the Explain phase, did learners spontaneously use the phrase 'undiminished in all directions' when describing the jerrycan, or did they need heavy scaffolding? If scaffolding was needed, the Model Update Statements will tell you whether the connection solidified by lesson end. (6) EQUITY CHECK — Were the same 4–5 learners dominating cold-call responses? Review your cold-call log and ensure Lessons 8 and 9 deliberately target learners who have been quiet. (7) EXTENSION LEARNERS — Did the matatu braking-system extension diagram reveal sophisticated understanding, or surface-level labelling? Use these to identify peer-teaching candidates for Lesson 8's group discussions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners pushed a small syringe plunger connected by water-filled tubing to a large syringe and observed the large plunger rise and lift a 200 g mass — a force greater than the input force. They also watched a hydraulic car lift in a Nairobi garage operate using the same principle at industrial scale.
What did I learn?	Pressure applied to a confined fluid is transmitted undiminished in all directions (Pascal's Principle). In a hydraulic system, $F_1/A_1 = F_2/A_2$, allowing a small force on a small area to produce a large force on a large area. This same principle explains the jerrycan collapse: the atmosphere transmitted its full pressure — undiminished, in all directions — onto every square centimetre of the jerrycan's surface the moment internal pressure was removed.
How does this explain the phenomenon?	Why hydraulic machines (car lifts in Changamwe, sugar presses in Mumias, hospital beds in Kenyatta National Hospital) can multiply force using only fluid and geometry — and why the crushed jerrycan is the inverse of a hydraulic machine: instead of internal pressure doing work outward, external atmospheric pressure did work inward, unopposed.

LESSON 8 (80 minutes): Pressure in Liquids: Deriving and Applying P = ρgh	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To derive the equation P = ρgh from first principles and apply it to calculate liquid pressure in Kenyan real-world contexts.
Knowledge	Students will know that pressure in a liquid depends on the density of the liquid, gravitational field strength, and depth; and that P = ρgh gives the pressure due to a liquid column alone.
Skills	Students will be able to: (a) derive P = ρgh from the weight of a liquid column; (b) substitute correct SI units and solve for pressure, depth, or density; (c) calculate liquid pressure at specified depths in Kenyan contexts (Sasumua Dam, Indian Ocean off Mombasa, Nairobi water tower).
Attitudes	Students appreciate that mathematical relationships in physics are not arbitrary — they emerge from physical reasoning — and that the same equation governs systems as different as a dam wall, a deep-sea diver, and a municipal water tower.
Key Inquiry Question	How does pressure in fluids affect the design of machines and structures?
Purpose in Storyline	In Lessons 1–7, students established qualitatively that liquid pressure increases with depth, depends on density, and acts in all directions. This lesson converts that qualitative understanding into a precise quantitative tool — the equation P = ρgh — so that students can, for the first time, put actual numbers on the invisible pressure that crushed the jerrycan and that engineers must account for when designing every Kenyan dam, pipeline, and water tower.
Safety Notes	No hazardous materials in this lesson. If using water-filled measuring cylinders during the derivation demonstration, ensure benches are dry to prevent slipping. Students handling calculators near water should keep devices away from any liquid. All waste water to be disposed of in a sink.

B. LESSON OVERVIEW
In this 8th evidence-gathering lesson, students move from qualitative understanding to quantitative mastery of liquid pressure. The lesson opens by reconnecting to the crushed jerrycan and asking: 'We know pressure increases with depth — but by HOW MUCH?' Students then derive P = ρgh from first principles by reasoning about the weight of a liquid column above an imaginary horizontal surface, building the algebra step by step with teacher guidance. They immediately apply the equation to three anchored Kenyan calculations: (1) pressure at the base of the Sasumua Dam reservoir (depth ≈ 40 m), (2) pressure on a diver at 10 m depth in the Indian Ocean off Mombasa (using seawater density 1025 kg/m³), and (3) pressure at the base of a Nairobi water tower (height ≈ 25 m). Students update their cumulative pressure models with quantitative labels, and the Driving Question Board is updated to reflect: 'Now we can calculate the invisible push — what does that number mean for the jerrycan?'

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Watch Me Source: TED-Ed	Students are shown three images projected on the board: (A) the Sasumua Dam wall in Nyandarua County, (B) a scuba diver	BEFORE showing images, teacher says: 'Do not Google anything. I want your gut physics.' [Display the three Kenyan images.]	Prediction Anchoring with Cognitive Conflict: By asking students to commit to a number BEFORE instruction, the teacher	Observable indicator: Every student has a written estimate with reasoning before partner discussion begins. Teacher scans

Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	descending in the Indian Ocean off Mombasa, and (C) a tall Nairobi City Water & Sewerage Company water tower above Westlands. Beneath each image is a prompt card: 'Estimate the water pressure pushing on the base/wall/diver. Is it more or less than atmospheric pressure (~100,000 Pa)? How many times more?' Students write individual estimates in their exercise books with brief reasoning, then share with a shoulder partner. The class hears 4–5 predictions read aloud before any instruction begins. Finally, the teacher re-projects the crushed jerrycan image: 'By the end of this lesson, you will be able to calculate the exact pressure that would have acted on the jerrycan walls if it had been submerged in the Sasumua reservoir. Let's find the equation that gives us that power.'	Teacher reads each prompt aloud, then says: 'Write your estimate AND your reasoning — one sentence minimum. You have 3 minutes. Go.' [Circulate, read over shoulders, note misconceptions — particularly students who write 'pressure equals weight' without considering area, or who guess atmospheric pressure is irrelevant to liquids.] After 3 minutes: 'Turn to your partner. Compare estimates. Do you agree? Argue if you don't. 90 seconds.' [Cold-call 5 students — intentionally choose one high estimate, one low estimate, one student who mentioned density, one who mentioned depth, one who mentioned area.] Record all five predictions verbatim on the board under the heading 'Our Predictions — Lesson 8.' Say: 'We will return to each of these at the end of the lesson and mark who was closest — and more importantly, who was reasoning correctly even if the number was off.' WAIT TIME after each cold-call: minimum 12 seconds before moving on.	activates prior knowledge and creates a personal stake in the answer. Displaying predictions publicly ensures students are motivated to reconcile their estimate with the derived result — this is a proven strategy for deepening retention of the equation. Connecting predictions to the jerrycan phenomenon maintains storyline coherence.	for: (a) students who write only a number with no reasoning — prompt them with 'What physics did you use to get that?'; (b) students who conflate pressure with force — flag for targeted questioning during derivation; (c) students who correctly identify depth and density as key variables — these students will be called on first during the derivation to articulate their reasoning.
Observe Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher guides the class through a first-principles derivation of P = ρgh using a step-by-step Socratic sequence. A large diagram is drawn on the board: a rectangular column of liquid of cross-sectional area A and height h sitting on a horizontal surface inside a larger container of liquid. Students copy each step into their books as it is built, filling in blanks on a structured derivation worksheet. Step 1 — Mass of the column: m = ρ × V = ρ × A × h. Step 2 — Weight of the column (gravitational force): W = mg = ρAhg. Step 3 — Pressure at the	Begin by drawing only the column outline on the board. Say: 'I have drawn a column of liquid. It has a cross-sectional area — let's call it A. It has a height — let's call it h. The liquid has a density — let's call it ρ. My question to you: what is the WEIGHT of this column pressing down on the base?' [WAIT 12 seconds. Cold-call 2 students.] After students suggest W = mg, say: 'Good. Now I need mass. How do I get mass from density, area, and height?' [WAIT 12 seconds. Cold-call 2 students.] Build each algebra line only AFTER students have supplied the	Socratic Step-Build Derivation: Rather than presenting the equation ready-made, the teacher elicits each algebraic step from students through targeted questions. This strategy ensures students understand the equation as a consequence of physical reasoning (weight, area, density) rather than a formula to memorise. The structured worksheet with blanks forces active engagement. The verification demonstration provides empirical confirmation of the algebraic result, linking mathematical and experimental evidence — a core NGSS Science	Observable indicator: Students can correctly fill in all five blanks on the derivation worksheet without being told the answers — only after supplying each step verbally. Teacher checks: (a) Does the student write 'V = Ah' correctly, or do they write V = A + h (additive confusion)? (b) Does the student identify the cancellation of A without prompting? (c) Can the student state in words: 'Pressure in a liquid depends only on density, gravitational field strength, and depth — not on the volume or shape of the container'? Students who cannot state point (c) receive

	base of the column: $P = F/A = \rho Ahg / A = \rho gh$. Students circle the cancellation of A and write in red: 'Pressure due to a liquid column is INDEPENDENT of the cross-sectional area.' The teacher then runs a quick verification demonstration: a tall measuring cylinder (50 cm) and a short wide beaker are both filled with water to the same depth. A rubber pressure sensor (or a simple manometer made from rubber tubing and food colouring) is placed at the base of each. Students observe that the pressure readings are equal, confirming that area does not matter — only depth and density.	reasoning. Do NOT write the next line until a student has said it aloud. When A cancels: 'Something beautiful just happened in the algebra. What disappeared?' [WAIT 15 seconds — this is the key moment. Cold-call 3 students.] After students identify that A cancels: 'Say that in physics words, not algebra words.' [WAIT 12 seconds.] Expected answer: 'The area doesn't matter — pressure only depends on depth and density.' Write on board in a box: $P = \rho gh$. Say: 'This is one of the most powerful equations in this unit. It tells us the pressure at any depth in any liquid — anywhere on Earth.' For the verification demo: 'Before I show you the result, predict: will the pressure at the base of the tall cylinder be more than, less than, or equal to the pressure at the base of the short beaker — given they are the SAME depth?' [Collect predictions by show of hands. Record on board.] Run demonstration. Say: 'What does this tell us about the jerrycan? Does it matter whether the jerrycan is tall and thin or short and wide?' [WAIT 12 seconds. Cold-call 2 students.]	and Engineering Practice (Using Mathematics and Computational Thinking; Planning and Carrying Out Investigations).	a targeted re-explanation during the application phase.
Explain Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES	Students work through three structured calculation problems in pairs, each set in a Kenyan context. A printed or projected problem card is provided for each scenario. PROBLEM 1 — Sasumua Dam (Nyandarua County): 'The Sasumua Dam supplies water to Nairobi. The reservoir depth near the dam wall is approximately 40 m. Calculate the pressure exerted by the water at the base of the dam wall.	Distribute problem cards (or project each problem one at a time). Say: 'Work with your partner. You have 5 minutes per problem. Write every step — do not skip the substitution line. Go.' [Circulate actively. For pairs who are stuck, ask: 'What is P equal to? What are ρ, g, and h for this problem? Now substitute.' Do NOT give the answer — only redirect to the equation.] After 5 minutes per problem, say: 'I need a volunteer	Contextualised Problem Solving with Progressive Complexity: The three problems escalate in complexity (straightforward substitution → comparison with atmospheric pressure → engineering reasoning) while all using $P = \rho gh$. This scaffolded sequence ensures procedural fluency before conceptual synthesis. The synthesis question explicitly bridges the calculation back to the anchoring	Observable indicators: (1) In the substitution line, students use correct SI values — ρ in kg/m^3, g = 10 N/kg, h in metres — not mixed units. (2) Students include Pa (pascals) as the unit in every answer. (3) For Problem 2, students correctly compare their answer to 101,325 Pa and draw a meaningful conclusion. (4) For the synthesis question, students correctly rearrange $P = \rho gh$ to $h = P/\rho g$ (or use proportional

for similar readings	(Density of fresh water = 1000 kg/m³, g = 10 N/kg).' PROBLEM 2 — Indian Ocean Diver off Mombasa: 'A marine biologist from the Kenya Marine and Fisheries Research Institute (KMFRI) dives to a depth of 10 m in the Indian Ocean off Mombasa to study coral reefs. Calculate the pressure due to the seawater at this depth. (Density of seawater = 1025 kg/m³, g = 10 N/kg.) How does this compare to atmospheric pressure (101,325 Pa)?' PROBLEM 3 — Nairobi Water Tower: 'A water storage tower in Westlands, Nairobi holds water at a height of 25 m above the supply pipes at ground level. Calculate the pressure the water exerts at the base of the tower. (Density of fresh water = 1000 kg/m³, g = 10 N/kg.) Explain why engineers design water towers to be tall.' After pairs solve each problem, whole-class solutions are built on the board by student volunteers writing one line at a time. Students then answer a synthesis question individually: 'The crushed jerrycan had atmospheric pressure (≈100,000 Pa) acting on its outside walls. Using $P = \rho gh$, at what depth in the Sasumua reservoir would the water pressure EQUAL the atmospheric pressure that crushed the jerrycan?' (Answer: $h = P/\rho g = 100{,}000/10{,}000 = 10$ m.)	to write ONLY the substitution line on the board — not the answer, just the substitution.' [Call on a mid-level student.] Then: 'I need a different volunteer to write the answer with units.' [Call on a second student.] After each answer is written: 'Class — do we agree? Thumbs up, thumbs sideways, or thumbs down.' [Address any thumbs-down responses before moving on. WAIT 10 seconds for thumbs.] For Problem 2, after the answer (102,500 Pa): 'Compare this to atmospheric pressure. What do you notice?' [WAIT 12 seconds. Cold-call 3 students.] Expected insight: seawater pressure at 10 m is approximately equal to one atmosphere. Say: 'So a diver at 10 m depth has TWO atmospheres pressing on them — one from the air above the water, one from the water itself. Remember this — it connects to what crushed our jerrycan.' For the synthesis question, say: 'This is individual — no partner. 3 minutes. This is the question that links EVERYTHING we have done today back to our crushed jerrycan.' [Collect 3 books to check synthesis answers.]	phenomenon, fulfilling the NGSS practice of Constructing Explanations — students use a mathematical tool to explain the original puzzling observation. Naming the Kenyan institutions (KMFRI, Nairobi City Water & Sewerage Company) grounds the physics in students' lived national context.	reasoning) and arrive at h ≈ 10 m. Teacher collects 3 exercise books at random to check synthesis answers before the next phase; if more than one of the three has a wrong method, the class receives a 2-minute targeted re-explanation before Phase 4.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES	The class gathers around the Driving Question Board. The teacher reads aloud the current driving question: 'Why did the jerrycan crush itself — and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?' Students review the sticky notes	Stand at the DQB. Say: 'Before you write your sticky note, look at everything we have put on this board over the past seven lessons. We have been building a story. Today we added something new — a number. Numbers change everything in physics.' [WAIT 15 seconds — allow students to read	Driving Question Board as Cumulative Narrative Tracker: The DQB is not a display — it is a living record of the class's growing understanding. By requiring students to either answer a prompt card quantitatively OR generate a new question, the teacher ensures both convergent thinking (applying	Observable indicators: (1) Factual answer sticky notes include a numerical answer with units — not just a qualitative statement. (2) New-question sticky notes are physically meaningful — they use the variables ρ, g, or h correctly in their phrasing, even if informally. (3) At least 80% of the class can

for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES for similar readings	from Lessons 1–7. The teacher posts three new prompt cards on the DQB under the heading 'Lesson 8 — Now We Can Calculate': (A) 'What is the pressure at 10 m depth in the Sasumua reservoir?' (B) 'Why must a Nairobi water tower be tall, not just wide?' (C) 'At what depth would the ocean water pressure equal the pressure that crushed our jerrycan?' Students individually write one new sticky note to add to the DQB. The sticky note must either: (a) answer one of the three new prompt cards using P = ρgh, OR (b) pose a new question that the equation raises for them (e.g., 'If seawater is denser than fresh water, does the equation predict fish feel more pressure in Lake Victoria than in the Indian Ocean at the same depth?'). Five sticky notes are read aloud. The teacher explicitly connects each note to the overall driving question narrative: 'Every number we can now calculate represents a real, invisible force — the same kind of force that slowly, relentlessly, crushed our jerrycan.'	the board silently.] Say: 'Write your sticky note. It must either answer a prompt card using P = ρgh, or it must ask a new question that the equation makes you wonder about. You have 2 minutes.' [Circulate. If a student is stuck: 'Look at the synthesis question you just solved. Can you put that answer on a sticky note?'] After 2 minutes: 'Come up and post your note.' [While students post, teacher reads notes quietly and selects 5 that represent a range — one factual answer, one insightful comparison, one new question, one misconception to address, one excellent connection to the phenomenon.] Cold-call the authors of those 5 notes to read and explain them. For the misconception note, say: 'This is a really interesting idea — let's examine it together' [do NOT say 'that's wrong']. After all 5 are discussed: 'Our driving question is getting smaller. We can now calculate the invisible push. Next lesson, we ask: what happens when that push is transmitted through pipes — and how does Kenya use that to save lives?'	the equation) and divergent thinking (extending the model). This dual requirement prevents the DQB from becoming a rote exercise. Reading selected notes aloud and discussing them publicly models scientific discourse — a core NGSS practice (Engaging in Argument from Evidence; Obtaining, Evaluating, and Communicating Information).	verbally connect their sticky note to the driving question when cold-called. Teacher flags any sticky note that suggests a fundamental misconception (e.g., 'deeper water is heavier so it pushes up, not down') for individual follow-up at the start of Lesson 9.
Model Building Phase 📹 **VIDEO:** Watch Me Source: TED-Ed Lessons 🔍 Search ARES for similar videos 📖 **HTML:** Pressure versus Temperature (PLIX) Source: CK-12 🔍 Search ARES	Students retrieve their cumulative pressure model — the cross-sectional diagram of a liquid container they have been updating since Lesson 4. In previous lessons, they added qualitative arrows showing pressure direction and relative magnitude (longer arrows = more pressure at greater depth). Today's task: add QUANTITATIVE LABELS to the model. Each student selects ONE of the three Kenyan contexts from the Explain phase (Sasumua Dam / Indian Ocean diver / Nairobi	Distribute or instruct students to open their cumulative model pages. Say: 'Your model has been growing with us since Lesson 4. Today it grows up — it gets numbers. Choose ONE of our three Kenyan contexts. Redraw and relabel. You have 15 minutes.' [Write the four requirements on the board and leave them there throughout.] [Circulate with targeted questions: 'What depth are you drawing your second line at? What does P = ρgh give you there? Is your arrow longer or	Proportional Arrow Scaling on Cumulative Models: Requiring students to draw arrows whose lengths are proportional to calculated pressure values forces a deep integration of mathematical and visual/spatial reasoning. The scale requirement (e.g., '1 cm = 50,000 Pa') is a scientific modelling practice — students must think about representation, not just calculation. The gallery walk adds a peer-review dimension: students must critique and be critiqued using physics	Observable indicators: (1) Pressure values at all three depth lines are numerically correct (teacher checks at least 8 models during circulation). (2) Arrow lengths are visibly proportional — the deepest arrow must be clearly longer than the shallowest; accept ±10% proportional error. (3) The scale is written and consistent. (4) The 'crushed jerrycan' sentence makes a specific, quantitative, or causal connection — not just 'pressure is big' but something like 'at 40 m depth, the pressure

for similar readings	water tower) and redraws their model to represent that specific context. Requirements for the updated model: (a) Label the liquid, its density in kg/m³, and the value of g. (b) Draw and label at least THREE horizontal depth lines (e.g., surface, mid-depth, maximum depth), with the calculated pressure value in Pa written at each level using $P = \rho gh$. (c) Draw pressure arrows — their LENGTH must be proportional to the pressure value (students must show their scale, e.g., '1 cm = 50,000 Pa'). (d) Write one sentence at the bottom of the model: 'This matters for the crushed jerrycan because...' A 'gallery walk' of 5 minutes allows students to view three peers' models and leave one sticky-note comment ('I notice...', 'I wonder...', or 'I question...').	shorter than the arrow at the surface? Why?' For students who finish early: 'Add a fourth depth line. Add a note about what material the real structure (dam wall / diving suit / tower pipe) must be strong enough to withstand.'] After 15 minutes: 'Models down. Gallery walk. You have 5 minutes. Visit at least three different models — they may show different Kenyan contexts. Leave one sticky note comment on each.' [After gallery walk:] 'Return to your seat. Read the comments on your model. Does any comment make you want to revise anything? If yes, make the revision in a different colour.' [WAIT 2 minutes for revision.] Close by saying: 'Hold up your model. Look at the arrows at the deepest point. That force — right there, in newtons per square metre — is real. It acts on dam walls, on divers, on pipes, and it acted on our jerrycan. Next lesson, we discover what happens when we transmit that force deliberately — through a fluid — to lift a matatu engine or operate a hospital's life-support equipment.'	language ('I notice the arrow at 40 m should be four times longer than at 10 m'). The sentence 'This matters for the crushed jerrycan because...' closes every model revision by re-anchoring to the phenomenon — preventing the model from becoming an isolated exercise. This addresses NGSS practices: Developing and Using Models; Using Mathematics and Computational Thinking.	(400,000 Pa) is four times the atmospheric pressure that crushed the jerrycan, which is why the Sasumua Dam wall must be so thick.' (5) Sticky-note gallery comments use at least one physics term (pressure, depth, density, $P = \rho gh$, Pa).

D. TEACHER REFLECTION

After the lesson, consider: (1) DERIVATION SUCCESS — Did most students supply each algebraic step when asked, or did I end up telling them? If I told more than two steps, plan a re-derivation warm-up at the start of Lesson 9 using a different liquid (e.g., cooking oil, density 920 kg/m³) to check retention. (2) UNIT ERRORS — How many students wrote answers without 'Pa' as the unit, or mixed cm and m in substitutions? If more than 30% made unit errors, build a 5-minute unit-checking routine into Lesson 9. (3) SYNTHESIS QUESTION — Did students successfully rearrange $P = \rho gh$ to solve for h? Rearrangement is a prerequisite for Lesson 9 (Pascal's principle applications). If fewer than 70% got the synthesis question correct, address algebraic rearrangement explicitly before Pascal's principle. (4) DQB QUALITY — Were students generating new questions using the variables of $P = \rho gh$, or were questions still qualitative ('why does water push?')? Qualitative DQB questions after this lesson indicate the equation has not yet been internalised as a thinking tool. (5) MODEL PROPORTIONALITY — Were arrow lengths genuinely proportional, or did students draw arbitrary lengths? Non-proportional models suggest students can calculate $P = \rho gh$ but cannot translate numbers into a visual representation — a key modelling skill that needs explicit attention in model-building sessions. (6) KENYAN CONTEXT ENGAGEMENT — Did students show increased engagement when problems were set at Sasumua Dam and the Mombasa coast versus generic 'a container of water'? Note this for evidence when reporting on PCIs and contextualised learning to the HOD.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed a step-by-step derivation of $P = \rho gh$ from the weight of a liquid column, including a live demonstration confirming that pressure at a given depth is independent of the container's cross-sectional area (using a tall measuring cylinder and a short wide beaker filled to the same depth).
What did I learn?	Students learned that pressure in a liquid depends only on the liquid's density (ρ), gravitational field strength (g), and the depth (h) below the surface, expressed as $P = \rho gh$. They calculated specific pressure values at the base of the Sasumua Dam (~400,000 Pa), at 10 m depth in the Indian Ocean off Mombasa (~102,500 Pa), and at the base of a 25 m Nairobi water tower (~250,000 Pa).
How does this explain the phenomenon?	Students explained that the atmospheric pressure (~100,000 Pa) that crushed the jerrycan is equivalent to the pressure at just 10 m depth in fresh water. This means structures at greater depths — like the Sasumua Dam wall — experience multiples of the force that collapsed the jerrycan, which is why engineers must design these structures with materials strong enough to withstand calculated pressure values, not just rough estimates.

LESSON 9 (40 minutes): PUTTING IT ALL TOGETHER: Unit Synthesis, Model Finalisation, and Driving Question Board Resolution

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students synthesise all pressure concepts learned across the unit into a single coherent explanation of the crushed jerrycan phenomenon, demonstrating mastery of particle-level, equation-level, and application-level understanding.
Knowledge	Students will be able to: (1) state and apply the definitions and equations for pressure (P equals F divided by A), liquid pressure (P equals rho g h), and Pascal principle (F1 divided by A1 equals F2 divided by A2); (2) explain atmospheric pressure and Boyle Law (PV equals constant) in relation to the jerrycan collapse; (3) connect pressure principles to Kenyan applications including dams, hydraulic brakes on matatus, weather forecasting, and medical syringes.
Skills	Students will: construct a final multi-level explanatory model (particle, equation, application); synthesise evidence from all eight preceding lessons; present a coherent group explanation; resolve questions on the Driving Question Board; complete a reflective exit ticket demonstrating personal conceptual growth.
Attitudes	Students will demonstrate intellectual courage in revisiting and revising early naive models, appreciate the power of scientific explanation to resolve puzzling phenomena, and show pride in the growth of their understanding across the unit.
Key Inquiry Question	How is pressure applied in day-to-day life, and how does pressure in fluids affect the design of machines and structures?
Purpose in Storyline	This is the culminating lesson of the unit. Students return to the anchoring phenomenon — the crushed jerrycan — armed with nine lessons of evidence, equations, and models. Every concept introduced across the storyline (atmospheric pressure, liquid pressure, Pascal principle, Boyle Law, barometers, manometers, hydraulic applications) converges here into a single, complete, quantitative explanation. The Driving Question Board is formally resolved, marking the closure of the inquiry cycle.
Safety Notes	No new hazardous materials are introduced. If the teacher re-demonstrates the jerrycan collapse, ensure the jerrycan is placed on a stable surface away from students. Remind students not to use mouth suction on any tubing. Mercury barometer diagrams are used only as reference — no mercury is handled.

B. LESSON OVERVIEW

The 40-minute lesson is structured around five phases. Students first PREDICT by revisiting their Lesson 1 sketches and writing what they now think caused the jerrycan to collapse. They then OBSERVE a second live viewing of the crushed jerrycan (or a short video replay) with fresh eyes and new vocabulary. In the EXPLAIN phase, groups present their final cumulative models using a three-layer framework (particle level, equation level, application level). The class then RESOLVES the Driving Question Board together, classifying every original question as answered, partially answered, or leading to future inquiry. Finally, students complete a reflective MODEL BUILDING exit ticket. Throughout, the teacher uses structured wait time, targeted probing questions, and Socratic press to ensure depth of explanation rather than surface recall.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Video: Resistance in a Wire] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Atmospheric Pressure (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students receive back their original Lesson 1 prediction cards and force-diagram sketches. Working individually for 3 minutes, they write a revised prediction on a new card using the stem: 'I now think the jerrycan collapsed because...' They must reference at least two specific concepts and one equation from the unit. Students then share with a partner using a Think-Pair-Share protocol, comparing how their Lesson 1 language (likely 'the air got sucked out' or 'it got weak') differs from their current language. Pairs discuss: 'What was missing from your first explanation that you can now name precisely?' Selected pairs share with the class. The teacher posts three or four sample Lesson 1 cards on the board alongside new revised cards so the class can visually see conceptual growth. This activates all prior knowledge and creates cognitive readiness for the synthesis explanation.	Teacher distributes the original Lesson 1 prediction cards (kept from Lesson 1 by the teacher). Teacher says: 'Before we do anything else today, I want you to hold your Lesson 1 card. Read what the Grade 10 version of you wrote nine lessons ago. Do not judge it — just read it.' [WAIT TIME: 60 seconds of silent reading.] Teacher then says: 'Now, on a new card, write what you would say today. You have three minutes. Use at least two concept names and one equation. Go.' [WAIT TIME: 3 minutes of silent individual writing.] Teacher circulates, reading over shoulders, noting which students are still using vague language ('the air went out') and which are using precise language ('the internal gauge pressure dropped below zero relative to atmospheric pressure'). After pair-share, teacher cold-calls two pairs: one that shows strong growth and one that still has a gap. Teacher says: 'I notice Wanjiku used the word atmospheric pressure in her new card but not in her first one. Wanjiku, what is atmospheric pressure and why does it matter here?' [WAIT TIME: 8 seconds.] Teacher then bridges: 'Hold that thought — we are about to put the full explanation together as a class.'	Comparing Lesson 1 and Lesson 9 prediction cards makes conceptual growth visible and concrete. The Think-Pair-Share ensures every student activates prior knowledge before the group presentations begin. Posting the before-and-after cards on the board externalises the learning journey and builds collective ownership of the unit narrative.	Teacher reads revised prediction cards during circulation. Look for: (1) use of specific vocabulary (atmospheric pressure, pressure differential, Boyle Law, Pascal principle); (2) at least one equation referenced correctly; (3) particle-level language ('air molecules hitting the surface'). Students who use only vague language are flagged for targeted questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** [Video: Resistance in a Wire] Source: PhET Interactive Simulations (English)	The teacher places the original crushed jerrycan from Lesson 1 on the front demonstration table. Students observe it silently for 30 seconds. The teacher then asks: 'If I were to re-inflate this jerrycan — pump air back in through this nozzle — what do you predict will happen and why? Use the	Teacher holds up the crushed jerrycan and says: 'You have seen this before. But this time, you are not the same students who saw it in Lesson 1. Look at it again with everything you now know.' [WAIT TIME: 30 seconds of silent observation.] Teacher says: 'Before I touch it, write one	Re-observing the phenomenon after instruction (a Return to Phenomenon strategy) allows students to experience the shift from puzzlement to explanation. Watching the jerrycan both recover and re-collapse makes the pressure differential concrete and bidirectional, deepening	Teacher listens to spontaneous student comments during the re-observation. Listen for: (1) correct use of pressure differential; (2) reference to atmospheric pressure acting inward; (3) reference to Boyle Law during re-inflation. Common misconceptions to catch: students saying 'the air was

🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	language of pressure.' Students write a one-sentence prediction with an equation before the demonstration. The teacher uses a bicycle pump to inject air back into the jerrycan through a one-way valve. Students observe the jerrycan slowly pop back to its original shape. Students annotate their observation: 'The jerrycan recovered because...' The teacher then removes air again (using the hand pump) and students watch the collapse re-occur. Students annotate: 'The jerrycan collapsed because...' This re-observation with new vocabulary transforms the familiar phenomenon into a quantitative event rather than a surprising trick. Students are now ready to give a full causal explanation.	sentence: if I pump air back in, what happens and what equation tells you so?' [WAIT TIME: 45 seconds.] Teacher pumps air in slowly, narrating: 'I am increasing the internal pressure. Watch.' After the jerrycan recovers, teacher says: 'What equation just played out in front of you?' [WAIT TIME: 6 seconds.] Expect students to call out Boyle Law (PV = constant, as V increased, P ratio changed) and pressure balance. Teacher affirms: 'Yes — Boyle Law, and pressure equilibrium. Now I am removing air again. Watch the collapse. This time, I want you to narrate it in your head using pressure language.' [WAIT TIME: collapse takes 20 to 30 seconds — full silent observation.] Teacher says: 'What you just narrated in your head — that is what your group will present in the next phase.'	understanding of equilibrium. Asking students to narrate internally before presenting externally scaffolds the explanation task.	sucked in' (passive) rather than 'atmospheric pressure pushed the walls inward when the internal pressure dropped' (active, directional, causal).
Explain Phase 📹 **VIDEO:** Video: Resistance in a Wire Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups (same groups from previous lessons) have 8 minutes to finalise their cumulative model using the Three-Layer Model Framework poster provided: Layer 1 (Particle Level) — draw and annotate a cross-section of the jerrycan showing air molecule collisions on both surfaces before and after air removal; Layer 2 (Equation Level) — write and annotate the relevant equations: P equals F divided by A, P equals rho g h, PV equals constant, F1 divided by A1 equals F2 divided by A2, and the standard atmospheric pressure value of 101,325 Pa; Layer 3 (Application Level) — name two Kenyan applications from the unit (for example, Sasumua Dam, matatu hydraulic brakes, Kenya Meteorological Department barometer, Mombasa	Teacher distributes the Three-Layer Model Framework poster and says: 'You have 8 minutes. Your model must answer the driving question completely — not partly, completely. Every layer must connect back to the jerrycan. Go.' [WAIT TIME: 8 minutes of group work — teacher circulates, pressing groups with: 'What is the pressure difference in Pascals?', 'Show me where Boyle Law appears in your Layer 1 diagram.', 'Which Kenyan application uses Pascal principle and which uses Boyle Law?'] Teacher signals time. Groups present in sequence. After each group, teacher asks the class: 'Did they explain why the jerrycan walls moved inward — not just that they did?' [WAIT TIME: 4 seconds.] Teacher uses targeted press after each presentation: 'You	The Three-Layer Model Framework forces students to operate simultaneously at particle, mathematical, and application levels — the hallmark of deep scientific understanding. Public group presentations make explanations contestable and refinable. Building a class-canonical explanation together models how scientific consensus is constructed. Recording the canonical explanation on the board gives students an authoritative reference they authored themselves.	Teacher uses the Three-Layer poster as a rubric: Layer 1 assessed for particle-level accuracy (bidirectional pressure, net force direction); Layer 2 assessed for correct equation use with units; Layer 3 assessed for correct identification of which principle each application uses. Teacher notes groups that conflate Boyle Law and Pascal principle (a common confusion) for immediate in-flight correction during presentations.

	diver, Nairobi water tower, medical syringe) and explain which pressure principle each one uses. Each group then gives a 2-minute presentation. One member narrates Layer 1, one narrates Layer 2, one narrates Layer 3. The class uses a peer feedback protocol: two stars (two things the group explained clearly) and one question (one thing they want the group to clarify). Teacher records the best student-generated sentences on the board as the class builds a collective canonical explanation of the jerrycan phenomenon.	said the pressure outside was greater. Greater by how much? Can anyone put a number on that?' [WAIT TIME: 8 seconds.] After all groups, teacher synthesises the best elements into a class-canonical 5-sentence explanation written on the board collaboratively: (1) Atmospheric pressure at sea level is approximately 101,325 Pa, meaning air molecules exert this force on every square metre of any surface. (2) When the jerrycan was sealed and air was removed, internal pressure dropped below 101,325 Pa. (3) The pressure difference — Boyle Law: PV equals constant — meant the reduced air inside could no longer balance the outside push. (4) The net inward force on each side of the jerrycan wall equalled the pressure difference multiplied by the wall area (F equals delta P multiplied by A). (5) The same principle — pressure transmitted through fluids — explains the Sasumua Dam walls, matatu hydraulic brakes, and the Mombasa diver experience.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Resistance in a Wire Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric	The class Driving Question Board — maintained since Lesson 1 — is displayed at the front. The teacher reads aloud every question card posted across the nine lessons. Students vote (thumbs up, thumbs sideways, thumbs down) for each card: thumbs up means fully answered, thumbs sideways means partially answered or answered in a new way, thumbs down means still open or now pointing to future learning. For each card, the student who posted it (or any volunteer) states in one sentence how the unit answered it.	Teacher stands at the DQB and says: 'This board has carried our curiosity for nine lessons. Every question on here was asked by someone in this room. Let us honour them by answering them.' Teacher reads each card clearly. After each vote, teacher says: 'Who posted this? Can you answer it now?' If the student who posted it is unsure, teacher opens to the class: 'Who can help?' [WAIT TIME: 6 seconds before opening to the class.] Teacher makes a point of praising cards that led to rich inquiry: 'This question about	The DQB resolution ritual closes the inquiry cycle explicitly and publicly. Sorting cards into three columns models how science works — some questions get answered, some open new questions, some remain at the frontier. Celebrating new questions as a sign of growth reframes the end of a unit not as the end of curiosity but as the beginning of the next inquiry. Connecting unanswered questions to future sub-strands motivates continued engagement.	Teacher assesses oral responses as students answer their original question cards. Look for: (1) use of unit vocabulary in the answer; (2) ability to connect the answered question to the phenomenon; (3) quality of new questions generated (more specific and technical = higher order thinking). Teacher notes any persistent misconceptions surfaced during the DQB review for inclusion in the teacher reflection.

Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Cards are physically moved into three columns: ANSWERED, LEADS TO FUTURE INQUIRY, and NEW QUESTIONS THIS UNIT CREATED. The class identifies which new questions connect to upcoming sub-strands (for example, questions about floating objects connect to upthrust and Archimedes principle; questions about wind pressure connect to fluid dynamics in later physics). Students celebrate that some of their new questions are more sophisticated than their original ones — a sign of genuine scientific thinking. The driving question itself — 'Why did the jerrycan crush itself and how does the invisible push of fluids move mountains, save lives, and power machines across Kenya?' — is answered by a student volunteer reading the class-canonical 5-sentence explanation from the board.	whether our bodies feel atmospheric pressure — that led us to the Magdeburg hemisphere, to barometers, to the entire atmospheric pressure lesson. That is what good scientific questions do.' After all cards are sorted, teacher asks: 'Look at the NEW QUESTIONS column. Are these better questions than the ones you asked in Lesson 1?' [WAIT TIME: 8 seconds — expect students to recognise that their new questions are more specific and use technical vocabulary.] Teacher concludes: 'A scientist who ends a unit with better questions than they started with has done real science. That is what you have done.' Teacher invites one student to read the driving question aloud, then invites another to read the class-canonical 5-sentence answer. Teacher says: 'The jerrycan is no longer a mystery. You solved it.'		
Model Building Phase 📹 **VIDEO:** Video: Resistance in a Wire Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Atmospheric Pressure (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students complete the unit exit ticket individually in 5 minutes. The exit ticket has three parts: Part A (Explain) — 'In three sentences, explain why the jerrycan collapsed. Use at least two equations and the phrase pressure differential.' Part B (Apply) — 'A sealed plastic water bottle is taken from Nairobi (altitude 1,661 m above sea level, atmospheric pressure approximately 83,000 Pa) to Mombasa (sea level, atmospheric pressure 101,325 Pa). The bottle was sealed at Nairobi. What happens to it at Mombasa and why? Use PV equals constant in your answer.' Part C (Reflect) — 'Complete this sentence: One thing I can now explain about pressure that I could not explain in Lesson 1	Teacher distributes exit tickets and says: 'This is individual. No talking. Five minutes. Show me what you know — not what your group knows.' [WAIT TIME: 5 minutes of silent individual work.] Teacher circulates, ensuring students are writing equations with units and not just narrative. Teacher does not help or prompt during this phase — it is the summative formative checkpoint. After collection, teacher asks: 'Part C — who wants to share what they can explain now that they could not before? Raise your hand.' Teacher takes three to four responses, affirming each one and connecting it to the specific lesson it came from: 'Wafula says he can now explain why the Sasumua Dam	The three-part exit ticket assesses simultaneously at recall (Part A), transfer (Part B — a novel context students have not seen before), and metacognition (Part C). The Nairobi-to-Mombasa bottle question is a deliberate transfer task: it uses familiar concepts in an unfamiliar direction (pressure increasing rather than decreasing), testing genuine understanding rather than memorised procedure. The ceremonial close with the jerrycan creates an emotional anchor for the learning — students will remember this moment and the explanation attached to it.	Exit ticket Part A is assessed for: two correct equations with units, correct use of the phrase pressure differential, causal direction (outside pressure greater than inside pressure causes inward net force). Part B is assessed for: correct identification that the bottle will be compressed at Mombasa (higher external pressure), correct application of PV equals constant (P1V1 equals P2V2, so higher P2 means smaller V2). Part C is assessed qualitatively for specificity — vague responses ('I learned about pressure') are flagged for follow-up; specific responses ('I can now calculate the pressure on a diver at 15 m depth using P equals rho g h') indicate genuine conceptual

	is...' Students hand in their exit tickets. The lesson closes with the teacher holding up the original crushed jerrycan one final time and saying: 'This jerrycan is now your graduation certificate. You know exactly what happened to it and why. Take that knowledge everywhere — to every dam wall, every matatu brake, every syringe, every weather report, every deep breath of Kenyan air.' Students record the complete class-canonical explanation in their notebooks as the final entry of the unit.	wall is thicker at the bottom — that came from Lesson 4, P equals rho g h. Perfect.' Teacher closes with the ceremonial moment: holds up the jerrycan and delivers the closing statement verbatim as scripted. Teacher says: 'Physics is not about memorising formulas. It is about seeing the invisible. You now see pressure everywhere. Well done.' [Final WAIT TIME: 10 seconds of silent appreciation before dismissal.]		growth.

D. TEACHER REFLECTION

After the lesson, reflect on: (1) Did every group achieve all three layers in their final model — particle, equation, and application? If Layer 1 (particle level) was weak across groups, plan a bridging activity connecting macroscopic pressure to molecular collision frequency for future units. (2) Were students able to transfer the jerrycan explanation to the novel Nairobi-to-Mombasa bottle question on the exit ticket? If fewer than 70 percent succeeded on Part B, the concept of pressure equilibrium and its bidirectionality needs reinforcement in the upthrust sub-strand. (3) Which DQB questions remained genuinely unresolved? Use these as the opening provocation for the next sub-strand. (4) Did the ceremonial DQB resolution feel authentic or procedural? The quality of student responses when answering their own original question cards is the truest measure of whether the inquiry cycle created genuine intellectual ownership. (5) Note any student whose Part C response showed exceptional growth — consider nominating them to present a unit summary to another class or to the school science club, connecting to the KICD emphasis on community engagement and peer teaching.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed the crushed jerrycan collapse and re-inflate in real time, connecting the visible deformation to the invisible pressure differential between the atmosphere and the reduced internal air pressure, and recognised both Boyle Law and atmospheric pressure acting simultaneously.
What did I learn?	Students learned that the complete explanation of the crushed jerrycan requires integrating all unit concepts: atmospheric pressure (101,325 Pa at sea level), the pressure differential when internal pressure drops, Boyle Law (PV equals constant) governing the internal air, Pascal principle explaining why the force is distributed uniformly across all jerrycan walls, and P equals rho g h for any liquid-pressure context. They also learned that unanswered questions are not failures but the frontier of future inquiry.
How does this explain the phenomenon?	Students explained that the jerrycan collapsed because removing internal air reduced internal pressure below atmospheric pressure (101,325 Pa), creating a net inward force on every square centimetre of the jerrycan wall (F equals delta P multiplied by A); the same principles explain the Sasumua Dam wall design, matatu hydraulic brakes, the Mombasa diver experience, the Kenya Meteorological Department barometer, and the medical syringe — demonstrating that the invisible push of fluids is the foundational mechanism behind machines, structures, and life-saving tools across Kenya.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.